Asymmetric Intellectual Property and Follow-on Innovation: Evidence from Drug Repurposing

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Job Market Paper

October 2026

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Abstract

Theories of cumulative innovation treat an upstream patent as a gate. This assumes symmetric protection: follow-on innovators can protect their own inventions from imitation. I show that when they cannot, the upstream patent also acts as a shield, because imitating a licensed follow-on innovation infringes it. Longer upstream protection can then increase follow-on innovation. Drug repurposing exhibits this asymmetry. A molecule's patent covers every use, but new-use rights are narrow and hard to enforce. Using new molecular entities approved by the FDA between 1985 and 2024, and instrumenting effective exclusivity with FDA review time, I estimate that one more year of exclusivity raises expected post-approval indications by about 34 percent. The marginal effect vanishes beyond roughly 15 years. Originators and entrants both respond positively, as the shield predicts. The results point to a failure in the protection of new uses, not a case for longer patents.

JEL codes: I18, L65, O31, O34
Keywords: cumulative innovation, patents, market exclusivity, drug repurposing, generic entry

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1 Introduction

Most innovation builds on earlier innovation, so the protection granted to an original invention shapes the incentive to build on it. The standard view treats the upstream patent as a gate. Follow-on innovators must license it, and transaction costs can block development (Green and Scotchmer 1995; Bessen and Maskin 2009). This view rests on an implicit assumption of symmetry: original and follow-on innovations can exclude imitation equally well (Scotchmer 1991; Green and Scotchmer 1995; Chang 1995; O'Donoghue, Scotchmer, and Thisse 1998). Under symmetry, the upstream patent adds no protection to a follow-on innovation that its own patent already protects, so longer upstream protection can only tax or delay follow-on innovation. Protection need not be symmetric. This paper shows that when the original technology holds a broad, enforceable right and the follow-on holds a narrow one that is hard to enforce, the upstream patent is also a shield. Because imitating a licensed follow-on also infringes the upstream patent, that patent protects the follow-on from imitation, and longer upstream protection can increase follow-on innovation. I test this prediction in drug repurposing, where patent law and its enforcement make protection sharply asymmetric.

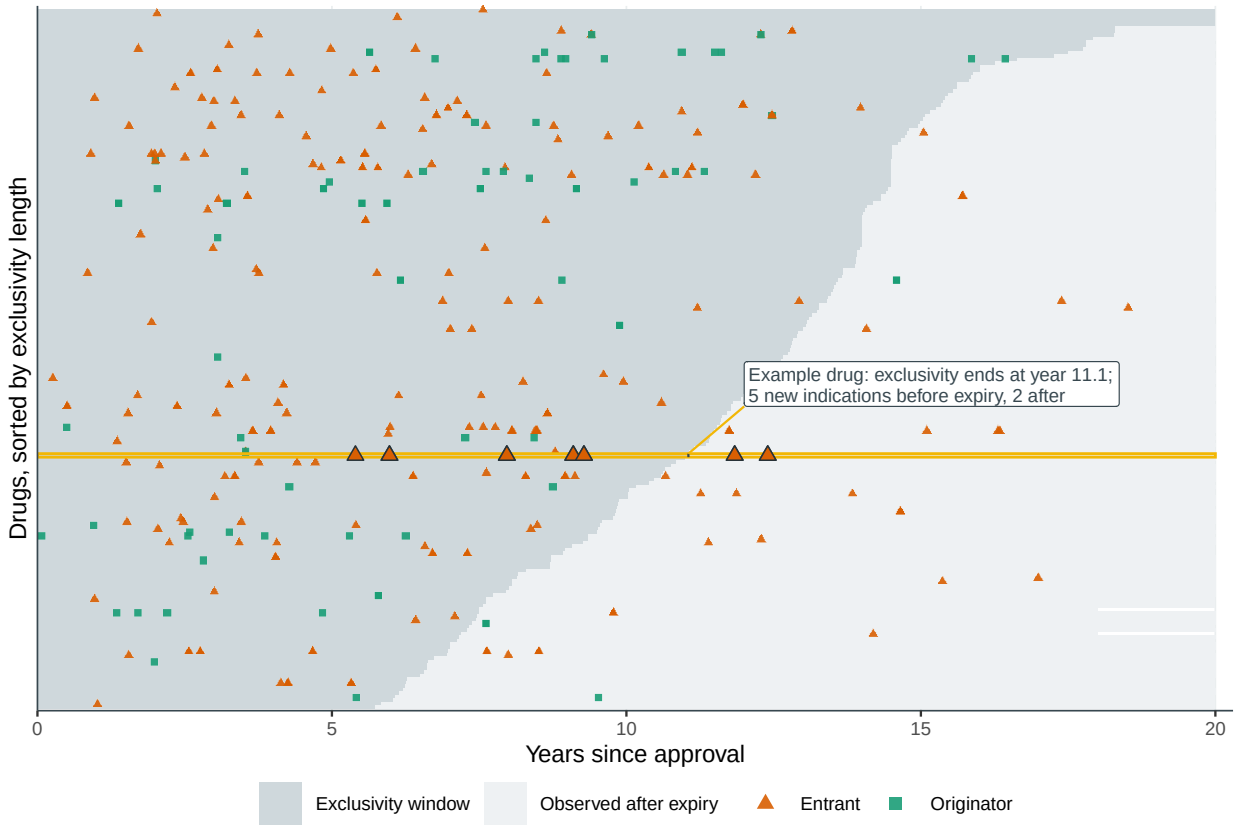
Drug repurposing applies an approved drug to a new disease. It is an important source of new treatments: nine of the twenty-six most transformative drugs of recent decades were repurposed from products developed for other purposes (Kesselheim, Tan, and Avorn 2015). Because the molecule and its safety profile are already established, repurposing is faster, cheaper, and less risky than developing a drug from scratch.[1] The new use also embodies the original molecule, so the molecule's patent covers it.

Figure 1 shows a pattern that the gate view does not predict. If a molecule's patent merely blocked follow-on innovators, its expiry should release new uses previously held back by licensing frictions. Instead, new indications cluster in the years before exclusivity expires, decline through expiry, and do not rebound afterward. Part of this decline reflects the depletion of a molecule's undiscovered uses. I ask whether the asymmetric structure of intellectual property rights also contributes.

In pharmaceuticals, the asymmetry has three sources. The first is claim scope. The patent on a drug's active ingredient claims the molecule itself, so it covers every use of the molecule, including uses unknown at filing. A patent on a new use covers only the increment. It cannot claim the molecule, which is prior art, and practicing the new use still infringes the molecule's patent. The second is enforcement. Once generics enter, physicians can

[1] See Ashburn and Thor (2004) and Pushpakom et al. (2019).

Figure 1: Timing of new indications relative to market exclusivity



Notes: 200 drugs drawn at random from the 337 approved before 2014, one per row, sorted by the length of market exclusivity. The dark bar runs from first approval to first generic entry; the light bar covers the years observed after generic entry; white lies beyond the observation period. Markers are new indications by drug age: triangles for entrants, squares for originators (the firm that owns the drug). The highlighted row is an example drug.

prescribe the molecule, pharmacists can substitute the generic, and the patentee cannot observe which indication a prescription serves; the value of a new use is information, which is hard to exclude (Eisenberg 2005; Roin 2014; Kapczynski and Syed 2013). Generic labels omit the patented indication, but physicians prescribe off label, and generic manufacturers are not liable unless they actively promote the patented use (Liddicoat et al. 2022). The third is regulatory exclusivity. A new molecule receives five years of data exclusivity covering the whole product, whereas a new indication receives three years, or seven for an orphan indication, covering only that indication (Langedijk et al. 2016). The molecule thus holds a broad, enforceable right, while the new use holds a narrow one that generics circumvent.

With asymmetric protection, the upstream patent plays two roles. It is a gate, because follow-on innovators need a license. It is also a shield, because imitating a licensed follow-

on uses the molecule and therefore infringes the upstream patent. Under symmetry the shield is redundant, since the follow-on's own patent already excludes imitation; under asymmetry it is the follow-on's main protection. An additional year of upstream protection therefore raises the value of licensed development, but adds a year of delay for developers who cannot obtain a license.

Section 2 formalizes this trade-off. An incumbent holds a broad patent with T_1 years remaining, and a developer can create new-use products that embody the incumbent's technology but have weak protection of their own. Before investing, the developer negotiates a license, which succeeds with a probability that captures licensing transaction costs; if negotiation fails, entry waits until the upstream patent expires. Follow-on development rises with T_1 when licensing succeeds enough for protection to outweigh delay (Proposition 1). When each additional use costs more to develop than the last, the response is single-peaked: it rises while protection dominates and falls once delay does. The gate view is the special case of symmetric protection, in which upstream protection adds nothing and follow-on development weakly falls with T_1 (Corollary 1).

The model yields three tests of asymmetry. First, longer upstream exclusivity raises follow-on development when licensing is efficient, whereas under symmetry it cannot (Proposition 1). Second, comparing originators with entrants isolates the shield (Proposition 3). An originator repurposing its own molecule needs no license, so it faces the shield but not the gate. Under symmetry, originator repurposing does not respond to T_1 and entrant repurposing does not rise with it. Under asymmetry, originator repurposing rises with T_1, and entrant repurposing also rises when licensing is efficient, though by less. Third, the more asymmetric the protection, the larger the effect of upstream protection (Proposition 2). I examine this prediction along two dimensions. By product form, a new indication added to the same product can be served directly by generics, whereas a new dosage form is protected by formulation patents and a combination adds an ingredient that single-molecule generics cannot replicate. By orphan status, rare-disease indications receive seven years of indication-level exclusivity rather than three.

To test these predictions, I construct a new drug-repurposing dataset covering all new molecular entities first approved by the FDA between 1985 and 2024. A repurposing event is measured as a post-approval grant of indication-level exclusivity. Effective market exclusivity is the interval from first approval to generic entry, measured from realized entry where observed and otherwise predicted from the drug's primary patent in the FDA's Orange Book. For each event, I record the approval date, orphan status, product form, and sponsor, tracing subsidiaries, parent firms, and acquisitions to classify the event as

originator or entrant repurposing.

Effective exclusivity is not randomly assigned, and three problems complicate estimation. Valuable molecules attract more patents and more patent challenges by generic firms, which shift exclusivity length, so I control for the patent stock at approval, time on the market, and therapeutic area. Exclusivity is measured with error when generic entry is unobserved. And firms that anticipate new uses may develop them before approval, which removes post-approval events from recent cohorts. Measurement error and anticipation both bias a naive estimate toward zero.

I instrument effective exclusivity with FDA review time, the interval from drug application's submission to approval. Patent term extension restores lost review time only up to a five-year cap, so where the cap binds or exclusivity ends at generic entry, each year of review consumes a year of post-approval protection. In the data, an additional review year reduces exclusivity by about half a year. The variation reflects FDA capacity rather than application characteristics: reviews shortened sharply after user fees funded more reviewers in 1992 and shifted again when statutory review goals changed in 1997 and 2012. The exclusion restriction requires review time to be unrelated to repurposing potential, conditional on controls. The main threat is clinical importance, because promising drugs receive priority review; the estimate barely changes when I control for priority review and remains positive among standard-review drugs. An alternative instrument, the interval between patent priority and the start of clinical trials (Gilchrist 2016), yields a similar estimate for the 220 molecules where it is observed, and an overidentification test does not reject.

An additional year of effective exclusivity raises expected post-approval indications by about 34 percent, or roughly 0.5 indications for the average molecule. The marginal effect is largest at short durations and indistinguishable from zero beyond roughly fifteen years, consistent with protection dominating at short horizons and licensing delays offsetting it at long ones.

Further results speak to asymmetry directly. Originator repurposing rises by about 72 percent per year of exclusivity, significant at the 10 percent level, although a firm that needs no license would not respond under symmetric protection. Entrant repurposing rises by about 25 percent, significant at the 5 percent level, although the gate view predicts no increase. The originator response is larger, as the model predicts, but the difference is not statistically significant. The effect is also present where asymmetry is starkest: new indications added to the same product, which generics substitute directly, rise by about 29 percent per year, significant at the 1 percent level. New dosage forms and combinations, which carry more protection of their own, have larger but insignificant point estimates.

Finally, rare-disease indications respond similarly to others despite seven-year orphan exclusivity, consistent with indication-level exclusivity doing little to offset the asymmetry, since it does not reach off-label generic use.

These results locate the policy problem in the asymmetry rather than in patent length. Upstream exclusivity promotes repurposing because the right on a new use is too weak to exclude imitation on its own. The response of repurposing to exclusivity therefore measures a failure of new-use protection, not a reason to lengthen molecule patents, and lengthening them would add little because the marginal effect vanishes beyond about fifteen years. Remedies should target the asymmetry: making the return to a new use enforceable, for example through differentiated prices for new uses, or replacing that return with public funding for repurposing trials. The mechanism is not specific to drugs: the shield operates wherever a follow-on innovation embodies an upstream technology but receives narrower or less enforceable rights than that technology.

This paper contributes to two literatures. First, it introduces asymmetric protection into models of cumulative innovation. Since Scotchmer (1991) and Green and Scotchmer (1995), these models have treated upstream patents as gates and asked how upstream breadth and length should divide profits between generations (Scotchmer 1996; Gallini and Scotchmer 2002). Scotchmer (1996) comes closest to relaxing symmetry: an exclusive upstream license can secure a second-generation monopoly even when the follow-on holds no patent of its own. Protection there remains complete while the upstream patent lasts, and the question remains how to divide profits. Here the follow-on's own right is weak, so the upstream patent supplies protection the follow-on would otherwise lack, and the gate result becomes the symmetric special case. O'Donoghue, Scotchmer, and Thisse (1998) show that length and breadth within one patent jointly protect an incumbent; here the rights that work together belong to different generations. The shield also helps explain why empirical findings differ across settings. Strong upstream rights and licensing frictions reduce follow-on innovation in several settings (Huang and Murray 2009; Williams 2013; Murray et al. 2016; Gaessler et al. 2025), but gene patents have no measurable effect (Sampat and Williams 2019), and blocking effects are concentrated in complex technology fields rather than pharmaceuticals (Galasso and Schankerman 2015). Where follow-on rights are weak, the shield offsets the gate. In the closest empirical study, Gilchrist (2016) finds that a first-in-class drug's exclusivity raises subsequent entry of substitutes by supporting their prices; here exclusivity raises development of complements by protecting them from imitation.

Second, the paper provides causal evidence on the problem of new uses documented in legal scholarship, which argues that new-use rights cannot sustain repurposing once

generics exist (Eisenberg 2005; Roin 2014; Kapczynski and Syed 2013). The results indicate that upstream patents supply the protection that new-use rights lack, and the response of repurposing to exclusivity measures the size of that shortfall. This adds an innovation margin to studies of generic entry, which find that originators shift promotion to alternative products still under patent protection once generic drugs enter (Lakdawalla and Philipson 2012; Castanheira, Ornaghi, and Siotis 2019). Generic entry lowers prices for existing uses, and it also removes the protection that sustains the development of new ones.

The results have three bounds. The Orange Book lists only small-molecule products, so biologic repurposing is excluded. The counts include only new uses that reach an FDA label, not trials by non-profit sponsors that do not seek approval, which may continue after generic entry. And the instrument shifts exclusivity only when patent term extension does not fully restore lost review time, so the estimate is local to those drugs.

Section 2 develops the model. Section 3 describes the institutional setting, Section 4 the data, and Section 5 the empirical strategy. Section 6 presents the results, Section 7 checks them against an alternative instrument, and Section 8 concludes. Proofs are in Appendix A, additional figures and tables in Appendix B, and further robustness checks in Appendix C.

2 The Model

I model a two-stage innovation process: a first innovation, and follow-on innovations that build on it. A follow-on innovation adapts the first technology to a new application. I ask how the *vertical structure* of intellectual property rights shapes how much follow-on innovation is developed. By vertical structure I mean the first innovation's protection relative to the follow-on's.

2.1 Setup

2.1.1 Firms, the Supply of New Uses, and Demand

Firms. There are three types of firms: an incumbent, a potential developer, and imitators.

The *incumbent* holds a patent on the first-generation technology i, with T_1 years remaining at date zero. Its development cost is sunk.

The *potential developer* can adapt the technology into new applications. Each new application has its own development cost $x \geq 0$. The supply is a flow. The number of applications

whose own cost is at most x is

$$G_i(x) \;=\; \gamma_i\, x^{\eta}, \qquad \gamma_i > 0,\ \eta > 0. \tag{1}$$

G_i is a power-function supply of new uses (Arora, Ceccagnoli, and Cohen 2008). In logs, $\ln G_i = \ln \gamma_i + \eta \ln x$: γ_i governs the level of new uses and η the elasticity of supply with respect to the value of a use. When $\eta < 1$ the supply is concave. Many new uses are low-hanging fruit, and each additional one is more expensive than the last. When $\eta = 1$ the supply is linear, and when $\eta > 1$ it is convex.

Imitators form a competitive fringe. They produce at zero marginal cost. They may supply any product whose quality existing products already embody, provided it infringes no patent in force. They cannot innovate beyond that frontier. Competition among them drives the price of whatever they may legally sell to zero.

Demand. The application markets are identical and independent, so it suffices to describe one. Demand is a flow. At every instant a unit mass of consumers, indexed by θ drawn from the exponential distribution with density $\frac{1}{m}e^{-\theta/m}$, decides whether to buy one unit at the current price. A consumer of type θ at every instant obtains

$$v + \theta\Delta - p$$

from the follow-on product, which adapts the technology to the new use, at price p, and zero from not buying. Here $v > 0$ is the value the first technology delivers in this market, and $\Delta > 0$ is the increment the follow-on product creates. The type θ is the marginal willingness to pay for that increment.

The original technology provides the same value to all consumers and is a known baseline at redeployment. Consumers differ in how they value the increment. The mean m measures its market value: a higher m scales up every consumer's willingness to pay. The exponential distribution captures the skewness typical of incremental improvements. Most consumers value the increment modestly. A small upper tail, such as severe cases in a disease population, finds it nearly indispensable.

I maintain throughout:

(A1) $m\Delta > v$; write $z \equiv v/(m\Delta) \in (0,1)$.

(A1) keeps the shielded monopoly price interior, so that some consumers are priced out. It is used only for the closed forms of Lemma 1 and the decomposition of Lemma 2. Without

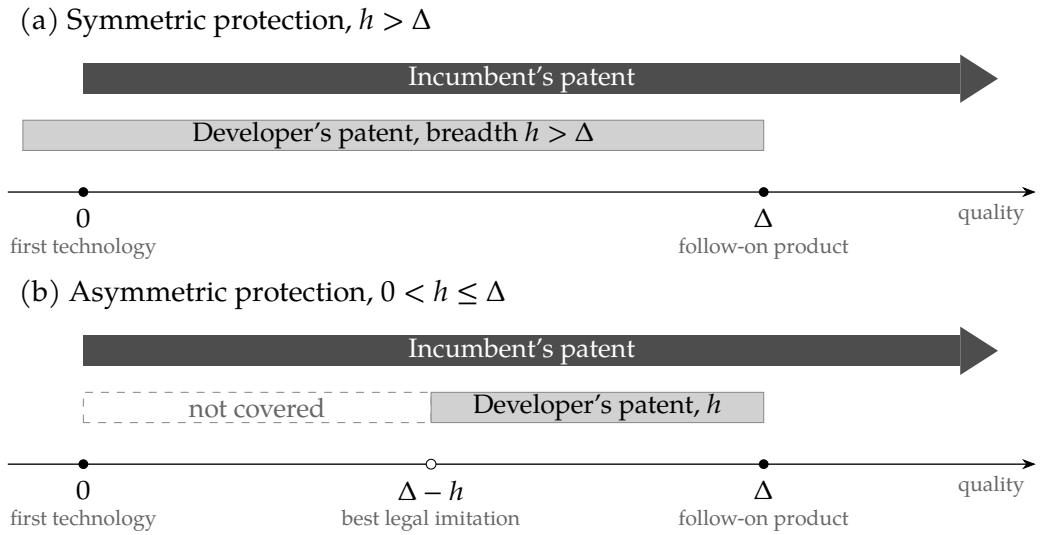
it the shielded price is v, the market is fully covered, and $\pi_S = v$. Every sign result below still holds, since $\pi_X = mh\,e^{-1} \leq m\Delta\,e^{-1} < v$.

2.1.2 Patent Breadth and the Vertical Asymmetry

O'Donoghue, Scotchmer, and Thisse (1998) distinguish two dimensions of patent breadth. *Lagging breadth* protects the patentee against lower-quality products, excluding imitations that copy or degrade the patented invention. *Leading breadth* protects it against higher-quality products, excluding later improvements built upon the patented technology. The two breadths work together with patent length to form effective market exclusivity, which lasts until a better product replaces the patented one.

Geometrically, a patent occupies an interval on the quality ladder. Lagging breadth extends that interval downward toward the prior art, leading breadth upward toward future improvements. I index the ladder by the increment a product embodies above the first technology, so the first technology sits at 0 and a follow-on product at Δ.

Figure 2: The incumbent's and the developer's patents on the quality ladder



Notes: Quality is the increment over the first technology, so the first technology sits at 0 and the follow-on product at Δ. The incumbent's patent covers every product that embodies its technology, below and above Δ. In panel (a) the developer's patent covers every product below its own quality. In panel (b) it covers only $(\Delta - h,\ \Delta]$, so once the incumbent's patent expires imitators may legally sell quality $\Delta - h$.

Patents in drug repurposing display this vertical asymmetry. A primary patent on an active ingredient corresponds to the incumbent's patent. Its lagging breadth covers the ingredi-

ent and every generic copy. Its leading breadth covers products built on the ingredient, including uses for a new disease. Leading breadth exceeding Δ is the empirically relevant case for the incumbent. The developer holds a secondary patent, typically a method-of-use patent on the new indication. It covers that indication for this technology, but not for other technologies.

Enforcement further limits secondary protection, so the enforceable increment, h, is much smaller than the increment created, Δ. Section 3.2 documents the problem: once generics exist, a new-indication patent cannot prevent the molecule from being dispensed for that indication, because physicians prescribe and pharmacists substitute at the molecule level and the patentee controls neither decision (Eisenberg 2005; Roin 2014; Kapczynski and Syed 2013).

Generic dispensing weakens protection for patented indications. In the United States, generics use Section viii "skinny labels" to omit patented indications. Pharmacists still substitute at the molecule level, and doctors prescribe off-label.[2] These omissions appear in many first generic approvals (Walsh et al. 2021), and generic sales extend to the omitted indications (Walsh et al. 2022). Cross-label[3] dispensing creates the same problem for second-medical-use patents (Liddicoat et al. 2021).

The model captures this asymmetry through patent breadth. The incumbent's leading breadth exceeds Δ, and its lagging breadth covers every imitation embodying its technology. The developer's patent has only lagging breadth h and cannot exclude other technologies serving the same use.

The parameter h[4] measures how much of the quality ladder the developer's patent protects, distinguishing the two panels of Figure 2. When $h > \Delta$, protection is symmetric: each patent covers every product below its own quality (panel a). The developer's patent then excludes imitation. When $0 < h \leq \Delta$ (panel b), it protects only the fraction h/Δ of the increment, leaving the rest exposed to imitation. It does not cover the baseline v, which the developer did not create and whose share of value is z. Lemma 2 shows that these two losses multiply. So $h < \Delta$ is the empirically relevant case for the developer, as leading breadth exceeding Δ is for the incumbent.

[2]The use of a drug for a condition that is not authorized for this drug.

[3]Using a generic drug to treat a disease for which another company has already obtained regulatory approval, where the generic's label omits this use because it is protected by a patent.

[4]h need not be a compact interval. It measures the competition that imitation can introduce.

2.1.3 Timing, Licensing, and the Development Rule

Time is continuous and runs on the developer's decision clock. At $\tau = 0$ all decisions are taken. Firms discount at $r > 0$, and I write

$$\begin{aligned} D(a,b) \;&\equiv\; \int_a^b e^{-r\tau}\, d\tau \;=\; \frac{e^{-ra} - e^{-rb}}{r}, \\ D(a,\infty) \;&=\; \frac{e^{-ra}}{r}. \end{aligned} \tag{2}$$

$D(a,b)$ is the date-zero value of a unit flow received over $[a,b]$. The value of a use is its profit flow times the D of the window in which that flow is earned.

The incumbent's remaining term T_1 divides the horizon into stage 1, $[0, T_1]$, while its patent is in force, and stage 2 thereafter.

Stage 1: the incumbent's patent governs the entire ladder.[5] The incumbent's patent blocks direct imitation through lagging breadth and all follow-on products through leading breadth. Developers therefore need its consent to enter any new-use market.

Licensing. At $\tau = 0$, before incurring any costs, the developer negotiates one license covering all new applications. Transaction costs, including asymmetric information, can prevent agreement (Scotchmer 1991; Chang 1995; Williams 2013; Galasso and Schankerman 2015; Gaessler et al. 2025). I model this risk in reduced form. Negotiation succeeds with probability ρ, permitting immediate entry for each developed use. It fails with probability $1 - \rho$, delaying all entry until T_1. I omit licensing fees because ex ante bargaining across independent markets makes fixed fees redistribute surplus without affecting development decisions.

For simplicity I assume:

> (**A2**) The developer's right on a use runs for T_2 years from the date that use is launched.
>
> (**A3**) Development costs are sunk at date zero, whatever the launch date.

Stage 2: only the developer's right remains. Once a use enters the market, its existence and supporting evidence become public. Imitators can reproduce any part of the increment that the developer's right does not protect. The best legal imitation therefore embodies

[5] For simplicity I assume the incumbent keeps its own product out of the new-use markets. Relaxing this does not change the signs in the propositions below.

$\Delta - h$. At a competitive price of zero, it delivers consumer surplus $v + \theta(\Delta - h)$. This exceeds v when $h < \Delta$ and equals v when $h \geq \Delta$.

Protection windows. A licensed developer has protection over $[0, T_2]$ for each developed use, overlapping the incumbent's patent during $[0, T_1]$. A blocked developer enters at T_1, with exclusivity over $[T_1,\ T_1 + T_2]$, entirely after the incumbent's patent expires. Let Π_{succ} and Π_{fail} denote the date-zero value of one developed use in the licensed and blocked branches, respectively. Sections 2.2 and 2.3 derive these values.

The development rule. The developer learns the licensing outcome at date zero, before investing. Application markets are identical and independent, so each application yields the same payoff. The developer develops an application if its cost is below that payoff. It develops $G_i(\Pi_{\text{succ}})$ uses when the licence is granted and $G_i(\Pi_{\text{fail}})$ uses otherwise. Averaging over these outcomes, the expected number of new uses developed is

$$n(T_1) \;=\; \rho\, G_i(\Pi_{\text{succ}}) + (1-\rho)\, G_i(\Pi_{\text{fail}}) \;=\; \gamma_i \left[\rho\, \Pi_{\text{succ}}^{\eta} + (1-\rho)\, \Pi_{\text{fail}}^{\eta}\right]. \tag{3}$$

2.2 Benchmark: The Gate View

Suppose that $h > \Delta$. Protection is symmetric: each patent covers every product below its own quality. The developer's own patent then blocks all imitation, including sale of the bare first technology, so no other firm may supply a new market during the developer's term T_2. The incumbent's patent matters only as a gate.

For each new application, the developer is therefore a monopolist throughout its window, whenever that window occurs. A consumer buys if $v + \theta\Delta \geq p$, so demand is $e^{-(p-v)/(m\Delta)}$. Maximising $p\, e^{-(p-v)/(m\Delta)}$ gives

$$\begin{aligned} p^* \;&=\; m\Delta, \qquad \text{coverage} \;=\; e^{z-1}, \\ \pi^* \;&=\; m\Delta\, e^{z-1}. \end{aligned} \tag{4}$$

A licensed developer earns π^* over $[0, T_2]$, so one developed use is worth $\Pi_{\text{succ}} = \pi^* A$, where

$$A \;\equiv\; D(0, T_2) \;=\; \frac{1 - e^{-rT_2}}{r}.$$

A blocked developer earns π^* over $[T_1,\ T_1 + T_2]$, worth $\Pi_{\text{fail}} \;=\; \pi^* D(T_1, T_1 + T_2) \;=$

$e^{-rT_1}\pi^* A$: the same profit, postponed. Substituting into the development rule (3),

$$n(T_1) \;=\; \gamma_i\,(\pi^* A)^{\eta}\Big[\rho + (1-\rho)\,e^{-\eta r T_1}\Big]. \tag{5}$$

Corollary 1 (The gate benchmark)**.**

$$\frac{\partial n}{\partial T_1} \;=\; -\,(1-\rho)\,\eta r\,\gamma_i\,(\pi^* A)^{\eta}\,e^{-\eta r T_1} \;\le\; 0,$$

with equality if and only if $\rho = 1$.

Proof. See Appendix A.

In the benchmark the incumbent's patent is a pure gate. The developer's own broad patent keeps imitators out, so any shield the incumbent's patent might offer is redundant. The blocked branch sees T_1 only as delay: the value of each use shrinks by e^{-rT_1}, and the count of uses worth developing shrinks by $e^{-\eta r T_1}$.

2.3 Shield: Asymmetric Protection

Now let the developer's protection be genuinely secondary:

(A4) $\quad 0 < h \le \Delta$; write $\kappa \equiv h/\Delta \in (0,1]$.

The developer's right on a use now spans two phases. In the *shield* phase $[0, T_1]$ the incumbent's patent excludes the fringe. In the *exposed* phase after T_1, consumers choose between the developer's product at price p and the best legal imitation, of quality $\Delta - h$ and free.

Lemma 1 (Prices)**.**

(*i*) *In the shield phase the price is* $p_S = m\Delta$*, coverage is* e^{z-1}*, and the profit flow is* $\pi_S = m\Delta\, e^{z-1}$*.*
(*ii*) *In the exposed phase the price is* $p_X = mh$*, the developer's market share is* e^{-1}*, the market is fully covered, and the profit flow is* $\pi_X = mh\, e^{-1}$*.*

Proof. See Appendix A.

Part (i) reproduces the monopoly problem in Section 2.2, with not buying as the outside option, so $\pi_S = \pi^*$. In part (ii), the consumer's choice depends only on the protected

increment h. Comparing the follow-on product with the free imitation cancels the baseline v and unprotected increment $\theta(\Delta - h)$. The exposed developer solves the pricing problem with quality h instead of Δ and a baseline of zero instead of v. The equilibrium price is $p_X = mh$, and market share is e^{-1}.

Lemma 2 (Rent retention)**.**

$$\phi \;\equiv\; \frac{\pi_X}{\pi_S} \;=\; \frac{h}{\underbrace{\Delta}_{\kappa}} \cdot \underbrace{e^{-z}}_{\textit{dilution}} \;\leq\; e^{-z} \;<\; 1.$$

Proof. See Appendix A.

The parameter ϕ measures the fraction of profit under the shield that survives the incumbent's patent expiration. Two factors determine this fraction. The enclosure ratio κ captures the loss from imitation: when imitators offer quality $\Delta - h$ for free, only the protected increment commands a positive price. The dilution factor e^{-z} captures the loss from a stronger outside option. Even with full protection ($\kappa = 1$), the free imitation supplies the baseline v, improving every consumer's outside option. A larger baseline share z leaves less monopoly rent. The complement $1 - \phi$ measures the share of profit supplied by the incumbent's shield.

The branch values follow. A licensed developer is shielded on $[0, T_1]$ and exposed on $[T_1, T_2]$. A blocked developer launches at T_1 and is exposed throughout $[T_1,\, T_1 + T_2]$. Hence

$$\Pi_{\text{succ}}(T_1) \;=\; \pi_S\, D(0, T_1) + \pi_X\, D(T_1, T_2), \tag{6}$$

$$\Pi_{\text{fail}}(T_1) \;=\; \pi_X\, D(T_1,\, T_1 + T_2) \;=\; e^{-rT_1} K, \tag{7}$$

where $K \equiv \pi_X A$. Substituting into (3) gives $n(T_1) = \gamma_i[\rho\, \Pi_{\text{succ}}^{\eta} + (1-\rho)\, \Pi_{\text{fail}}^{\eta}]$. From (6) and (7), $\partial \Pi_{\text{succ}} / \partial T_1 = (\pi_S - \pi_X)\, e^{-rT_1} > 0$: a marginal lengthening of T_1 shields the profit flow at that instant from generic erosion. For blocked developers, $\partial \Pi_{\text{fail}} / \partial T_1 = -rKe^{-rT_1} < 0$, which is the interest cost of postponing the fixed payoff K. Therefore

$$\begin{aligned} \frac{\partial n}{\partial T_1} \;&=\; \gamma_i \eta\, e^{-rT_1} B(T_1), \\ B(T_1) \;&\equiv\; \underbrace{\rho\, (\pi_S - \pi_X)\, \Pi_{\text{succ}}^{\eta-1}}_{\text{shield}} \;-\; \underbrace{(1-\rho)\, rK\, \Pi_{\text{fail}}^{\eta-1}}_{\text{gate}}. \end{aligned} \tag{8}$$

The term $\gamma_i \eta\, \Pi^{\eta-1} = G_i'(\Pi)$ denotes the density of marginal applications at Π: the number

a small rise in value brings in. The shield term is the probability of holding a licence, times the flow value of the shield, times the density at the licensed threshold. The gate term is the probability of being blocked, times the interest cost of postponing K, times the density at the blocked threshold. The sign of $\partial n/\partial T_1$ is the sign of B.

Proposition 1 (The sign of $\partial n/\partial T_1$)**.** *For every $\eta > 0$ and $T_1 \in [0, T_2]$, $\partial n/\partial T_1 > 0$ if and only if $\rho > \hat{\rho}(T_1)$, where*

$$\hat{\rho}(T_1) \;\equiv\; \frac{rK\,\Pi_{\text{fail}}^{\eta-1}}{rK\,\Pi_{\text{fail}}^{\eta-1} + (\pi_S - \pi_X)\,\Pi_{\text{succ}}^{\eta-1}} \;\in\; (0,1),$$

and $\hat{\rho}(0) = rK/(rK + \pi_S - \pi_X)$ for every η. Write $\rho_{\text{lo}} \equiv \hat{\rho}(0)$. Moreover:

(*i*) *$\eta = 1$: $\hat{\rho}$ is constant. So n is monotone in T_1: $\partial n/\partial T_1 > 0$ if $\rho > \rho_{\text{lo}}$, and $\partial n/\partial T_1 < 0$ if $\rho < \rho_{\text{lo}}$.*

(*ii*) *$\eta < 1$: $\hat{\rho}$ is strictly increasing in T_1. So $\partial n/\partial T_1$ changes sign at most once, from positive to negative, and n is single-peaked.*

(*iii*) *$\eta > 1$: $\hat{\rho}$ is strictly decreasing in T_1. So $\partial n/\partial T_1$ changes sign at most once, from negative to positive, and n is single-troughed.*

Proof. See Appendix A.

The incumbent's patent promotes follow-on applications when successful licensing is likely enough for protection to outweigh delay. Write $B = \rho\, a - (1-\rho)\, c$, where $a = (\pi_S - \pi_X)\Pi_{\text{succ}}^{\eta-1}$ measures the shield's benefit and $c = rK\,\Pi_{\text{fail}}^{\eta-1}$ measures the gate's cost. The derivative is positive when $\rho > c/(a+c) = \hat{\rho}(T_1)$.

At $T_1 = 0$, both branches have the same unprotected window worth K, so the marginal application densities cancel. The threshold then compares the shield's flow value with the interest cost of delay. As T_1 increases, the licensed value rises and the blocked value falls. The threshold's response depends on how marginal application density changes with value, governed by η. When $\eta < 1$, marginal application density falls with value, so the shield affects fewer marginal applications while the gate affects more. Gains shrink and losses grow, so $\hat{\rho}$ rises with T_1. Once ρ falls below the threshold, the derivative remains negative, yielding a single peak. When $\eta = 1$, marginal application density is constant, so $\hat{\rho}$ is unchanged. When $\eta > 1$, density rises with value, so $\hat{\rho}$ falls with T_1. A crossing then changes the derivative from negative to positive, yielding a trough.

Proposition 2 (Asymmetry as an amplifier)**.** *Under* (*A4*)*, for every* $\eta > 0$*,* $\rho \in [0,1]$ *and* $T_1 \in [0, T_2]$*,*

$$\frac{\partial^2 n}{\partial T_1 \, \partial h} \;=\; \frac{\gamma_i \eta \, e^{-rT_1}}{h}\Big[\rho \, a \, \varepsilon_a - (1-\rho)\, c \, \eta\Big],$$

$$\varepsilon_a \;\equiv\; \frac{\partial \ln a}{\partial \ln h} \;=\; -\frac{\phi}{1-\phi} - (1-\eta)\, s(T_1),$$

where $a = (\pi_S - \pi_X)\Pi_{\text{succ}}^{\eta-1}$ *and* $c = rK\,\Pi_{\text{fail}}^{\eta-1}$ *are the shield and gate terms of* (*8*)*,* $\phi = \pi_X/\pi_S$ *is the rent retention of Lemma 2, and* $s(T_1) = \pi_X D(T_1, T_2)/\Pi_{\text{succ}}(T_1)$ *is the share of a licensed use's value earned after the incumbent's patent expires, falling from* 1 *at* $T_1 = 0$ *to* 0 *at* $T_1 = T_2$*. The gate term is negative for every* η*.*

(*i*) *If* $\eta \le 1$*, then* $\varepsilon_a < 0$*, so the cross-partial is negative for every* ρ *and* T_1*.*

(*ii*) *If* $\eta > 1$*, then* $\varepsilon_a < 0$ *if and only if* $(\eta - 1)\, s(T_1) < \phi/(1-\phi)$*. Since* $s \le 1$*, the cross-partial is negative for every* ρ *and* T_1 *whenever* $\eta < 1/(1-\phi)$*.*

In the benchmark, h does not enter and the cross-partial is zero.

Proof. See Appendix A.

The incumbent's patent has the largest marginal effect when the developer's protection h is weakest. Increasing h raises standalone profit π_X, reducing the marginal gain a and increasing the marginal loss c from Proposition 1. The loss c always rises because delay postpones profit $K = A\pi_X$. Since π_X is proportional to h, the elasticity of c with respect to h is η. The gain a falls through two channels. Stronger protection reduces the shield flow $\pi_S - \pi_X$. When $\eta < 1$, the higher licensed value also lowers marginal application density. For $\eta \le 1$, stronger protection therefore reduces the gain and increases the loss. When $\eta > 1$, marginal application density rises, but the declining shield flow still dominates if $\eta < 1/(1-\phi)$. In the symmetric benchmark, π_X drops out, so h is irrelevant.

The developer can be the incumbent or an outside firm. I call the incumbent developing its own applications the originator and third-party developers entrants. The originator differs only in requiring no license, equivalent to frictionless self-licensing ($\rho = 1$). Both face the same application distribution (1) and earn π_S while the upstream patent lasts. After expiration, both earn π_X because the originator's secondary patent provides no broader protection than an entrant's. Segmented product markets leave primary-market profits unaffected by new-use decisions, so these profits drop out of the maximization problem.

The originator develops every project costing less than $\Pi_{\text{succ}}(T_1)$. The respective new application counts therefore satisfy

$$n_O(T_1) \;=\; G_i(\Pi_{\text{succ}}(T_1)) \;=\; \gamma_i\,\Pi_{\text{succ}}(T_1)^{\eta},$$
$$n_E(T_1) \;=\; n(T_1) \;\text{ of (3)}.$$

This distinction turns the licensing friction from an unobservable parameter into observable variation. Originator and entrant face identical technology, identical demand, and an identical timeline. They differ solely in who must obtain permission.

Proposition 3 (Originator and entrant responses)**.** *Under (A4), for every $\eta > 0$ and $T_1 \in [0, T_2]$,*

$$\frac{\partial n_O}{\partial T_1} \;=\; \gamma_i \eta\, e^{-rT_1}\,(\pi_S - \pi_X)\,\Pi_{\text{succ}}^{\eta-1} \;>\; 0,$$

$$\frac{\partial n_E}{\partial T_1} \;=\; \gamma_i \eta\, e^{-rT_1}\Big[\rho\,(\pi_S - \pi_X)\,\Pi_{\text{succ}}^{\eta-1} - (1-\rho)\,rK\,\Pi_{\text{fail}}^{\eta-1}\Big],$$

which is positive if and only if $\rho > \hat{\rho}(T_1)$ of Proposition 1. In the benchmark, $\partial n_O/\partial T_1 = 0$ and $\partial n_E/\partial T_1 \le 0$.

Proof. See Appendix A.

The originator benefits from upstream protection without needing a license. Because $\partial\Pi_{\text{succ}}/\partial T_1$ is positive, its development count rises with protection length for every η. The entrant faces both protection and licensing barriers, weighted by ρ and $1-\rho$. Its response has the sign of B in Proposition 1.

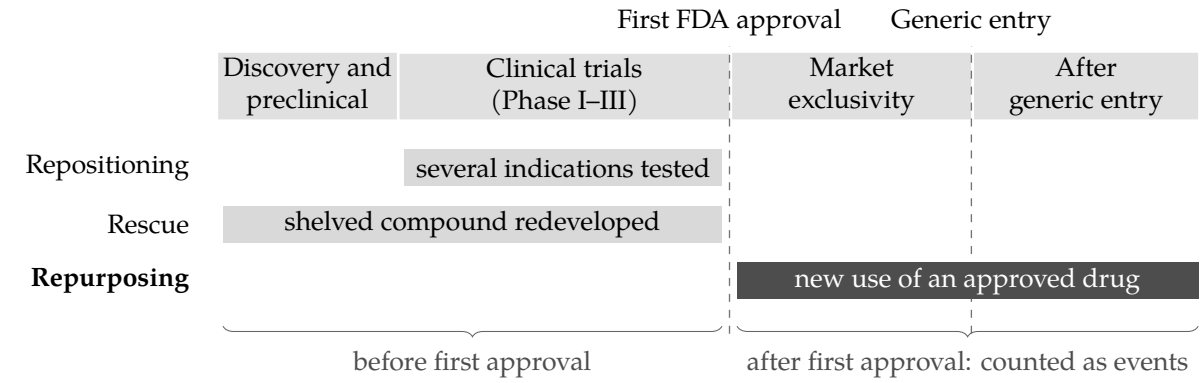
In the benchmark model, the developer's own patent fully excludes imitation. Upstream protection adds no shield, so the originator's development count is invariant to T_1. An observed originator response to exclusivity length is direct evidence of the shield effect. The entrant's response is strictly negative under the pure gate view but positive with the shield when the licensing market is efficient.

3 Institutional Background

3.1 What Counts as Drug Repurposing

The literature defines "repurposing" by timing, disease area, or technology (Murteira et al. 2014; Murteira, Millier, and Toumi 2014). I classify these activities by timing because I study follow-on innovation: a new use developed after the original innovation. This classification distinguishes three practices (Figure 3). Drug repositioning explores candidate indications before approval and advances the most promising to Phase III (Novac 2013). This exploration typically occurs in Phase II, after Phase I establishes the compound's safety. Drug rescue tests compounds that failed in development for alternative indications (Krishnamurthy et al. 2022). Both seek a molecule's first market entry, so neither constitutes follow-on innovation in the model. Drug repurposing seeks additional therapeutic uses after initial regulatory approval. It corresponds to a follow-on application in the model. Throughout this paper, a repurposing event is the post-approval addition of a new treatment indication.

Figure 3: Repurposing and related practices on the drug-development timeline



Notes: Drug repositioning tests several candidate indications within the original trials; drug rescue redevelops a compound shelved during trials for another indication. Both seek a molecule's first approval. Drug repurposing adds an indication to an approved drug, during market exclusivity or after generic entry. Only repurposing counts as an event in this paper.

A new indication reaches the market in several product forms, and the form matters for measurement; Table 1 lists the forms recorded in the data.

Table 1: Product forms of follow-on activity

Product form	Definition	Examples
Same molecule, new product	The same active ingredient approved as a separate product to treat a different disease	Bupropion: approved as Wellbutrin (1985) for major depressive disorder, and as Zyban (1997) for smoking cessation.
New dosage or formulation	The same molecule reformulated into a new dosage form, strength, or route	Finasteride: first approved as Proscar 5 mg tablets (1992) for benign prostatic hyperplasia; reformulated at a lower strength as Propecia 1 mg tablets (1997) for androgenetic alopecia (male pattern hair loss).
Combination	Two or more separately approved active ingredients combined into one product	Symbyax (olanzapine + fluoxetine), approved in 2003 for depressive episodes of bipolar I disorder—the first FDA-approved treatment for bipolar depression; its components were first approved for different diseases—fluoxetine (Prozac, 1987) for major depressive disorder, olanzapine (Zyprexa, 1996) for schizophrenia.

Two screens are applied to align the count with the repurposing definition. First, I retain only new, distinct therapeutic indications for already-approved active ingredients whose safety is established through prior clinical use. Each new use must have its own supporting safety and effectiveness data (Ashburn and Thor 2004; Pushpakom et al. 2019). I exclude active-ingredient variants, including salts, esters, ethers, isomers and mixtures of isomers, complexes, and other derivatives. These modify the active substance rather than its therapeutic use and share the parent drug's active moiety. Their development follows a

chemistry-driven route distinct from repurposing; see also Hancu and Modroiu (2022) on chiral switches. Second, I exclude line extensions, including broader patient populations within an approved indication. These expand an existing use rather than establish a distinct indication (Murteira et al. 2013). This exclusion is consistent with the FDA's definition. Its "new indication" refers to a new use supported by new safety and effectiveness data, rather than expansion of an approved indication.

3.2 Market Exclusivity of Drugs in the United States

Protection for an approved drug flows from two systems. The patent system supplies rights that run from the filing date and can be challenged or designed around; the regulatory system supplies exclusivities that run from FDA approval and can be neither invalidated nor infringed. The framework is the Hatch–Waxman Act of 1984, which answered two problems at once. Effective patent life was being lost to clinical development and review, and generic entry was delayed; the Act paired patent-term extension with data exclusivities and a 180-day reward for the first generic challenger. Targeted pull incentives sit alongside it, one predating it (orphan exclusivity, 1983) and the rest added later (pediatric extension, 1997; biologics, 2010; antibiotics, 2012). Table 2 summarises the instruments. Several rights typically attach to one product, and they do not stack symmetrically: patent-term extension and the pediatric extension lengthen the *patent* clock, while the remaining regulatory exclusivities extend only one another.

Table 2: Market exclusivity instruments in the United States

Instrument	Statute (year)	Runs from	Length	What it protects
Patent term	Patent Act; TRIPS (1995)	First filing	20 years	The claimed invention: product, use, or process
Patent term adjustment (PTA)	AIPA (1999)	Grant	Day-for-day for USPTO delay	The same patent
Patent term extension (PTE)	Hatch–Waxman (1984)	Approval, on request	Up to 5 years; effective life capped at 14	One patent per drug

Instrument	Statute (year)	Runs from	Length	What it protects
New chemical entity (NCE)	Hatch–Waxman (1984)	Approval	5 years	Data: no ANDA may rely on the originator's file
New clinical investigation (NCI)	Hatch–Waxman (1984)	Approval of the new use	3 years	Data supporting the new form or indication
Orphan drug exclusivity (ODE)	Orphan Drug Act (1983)	Approval with designation	7 years	Same drug, same indication
Pediatric exclusivity	FDAMA (1997)	Completion of requested studies	+6 months	Extends existing patents and exclusivities
Biologic reference product (BRP)	BPCIA (2010)	Approval	4 years (filing) / 12 years (approval)	Data: no abbreviated BLA
GAIN (QIDP)	GAIN Act (2012)	QIDP designation	+5 years	Extends non-patent exclusivities
180-day generic exclusivity	Hatch–Waxman (1984)	First commercial marketing	180 days	The first Paragraph-IV filer, against later ANDAs

Protection differs in breadth according to the innovation covered. A base patent broadly protects the active ingredient. Secondary patents cover alternative forms, formulations, dosages, and methods of use, providing narrow protection after the base patent expires (Abud, Hall, and Helmers 2015). Of 170 molecules approved in 1996–2004, only 12 relied solely on method-of-use patents for exclusivity (Hemphill and Sampat 2011). Among regulatory exclusivities, NCE and BRP bar reliance on originator data for the whole product. ODE and NCI protect only the approved indication (Langedijk et al. 2016).

A drug at first approval receives broad protection through a base patent and NCE or BRP. A repurposed indication can obtain only narrower protection: a secondary patent, NCI's three years, and orphan exclusivity if eligible. New-use protection is difficult to enforce once generics exist. Eisenberg (2005) explains that a new-indication patent cannot prevent dispensing for that indication. Physicians prescribe and pharmacists substitute generics;

the patentee controls neither decision. Roin (2014) identifies the enforcement problem: patentees cannot observe which indication a prescription serves and therefore cannot charge separately for the patented use. Kapczynski and Syed (2013) explain that a new use derives its economic value from information, which is difficult to exclude.

Off-label prescribing and cross-label substitution weaken protection for new uses. Off-label prescribing accounted for about a quarter of U.S. outpatient prescriptions in 2016–2021 (Blankart and Lichtenberg 2026). Cross-label substitution allows generics approved only for the original indication to serve the new one (Liddicoat et al. 2022; Belowska 2025). U.S. and European courts and statutes largely exempt generic firms from liability absent intentional promotion (Scheineson and Cuevas 2013; Van Norman 2023; Aboy et al. 2022). These practices make new-use exclusivity hard to enforce and reduce its economic rent. The stand-alone right therefore provides weak protection, corresponding to low h in panel (b) of Figure 2. The molecule's base patent provides effective protection for the repurposed indication until it expires.

4 Data

The unit of observation is the New Molecular Entity (NME), a product whose active moiety receives FDA approval for the first time. The sample includes all NMEs approved between 1985 and 2024. For each NME, I record repurposing events, patent and exclusivity protection, clinical development start, and initial disease area. I also identify the firms responsible for the original approval and each new indication.

4.1 Repurposing Events

I identify post-approval indications from their regulatory exclusivities in the Orange Book. These include new-clinical-investigation (NCI) exclusivity for changes supported by new clinical studies and orphan drug exclusivity (ODE). ODE covers uses treating conditions affecting fewer than 200,000 U.S. patients (Langedijk et al. 2016). NCI also covers changes in dosage form, strength, route, or patient population. Orange Book codes distinguish new indications (I-) from other changes (NDF, NS, NPP, D-, M- and others). I retain only I-coded NCI and ODE grants, both awarded only for new indications. Grants dated after first approval therefore count the molecule's post-approval follow-on applications, corresponding to repurposing events in the model. Each grant records an approval date

and sponsor.

The repurposing count is constructed in two steps. First, I extend the 1985–2016 Orange Book dataset from Durvasula et al. (2023) through 2024 to cover all regulatory exclusivities over 1985–2024. I parse the FDA's annual Orange Book editions, extract their text, and de-duplicate records across years. Second, I merge this panel with the NME population and retain I-coded NCI and ODE grants dated after each molecule's first approval. These grants yield the repurposing count per molecule used throughout.

The dataset covers product forms beyond same-molecule reformulations and dosage changes, including combinations, and dates each new application by approval. These features distinguish it from existing repurposing datasets (Akodad et al. 2026; Langedijk et al. 2016). The Orange Book's scope imposes two limitations. It lists only small-molecule products, excluding biologic repurposing. Section 4.6 explains why this restricts scope rather than creates selection bias. The data distinguish ODE from NCI grants, revealing orphan status but not the new indication's disease area. Neither limitation affects the tests below, which use counts rather than therapeutic content.

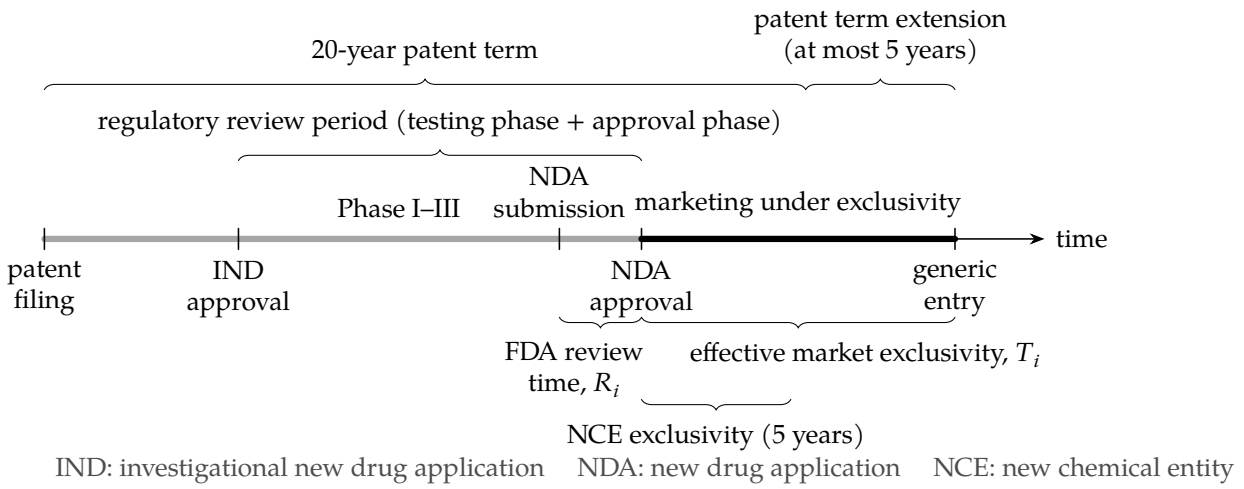
4.2 Patents and Effective Market Exclusivity

For each NME, I identify all patents and regulatory exclusivities granted between 1985 and 2024 in four steps. First, I construct the 1985–2024 Orange Book panel described above. Second, I link each molecule to its rights by merging the panel with the NME population. Third, I calculate protection periods using grant dates, expiry dates, and statutory lengths in Table 2, retaining rights in force at first approval. Fourth, I match associated patents to the USPTO's register of extended patents to identify patent-term extensions (PTE).

A drug's life runs from preclinical research through IND approval, the three clinical-trial phases, New Drug Application (NDA) review, marketing under exclusivity, and finally generic entry. *Effective market exclusivity* is the period between first NDA approval and the entry of the first generic competitor, the empirical counterpart of the model's T_1 (Figure 4).

For molecules with observed generic entry, I measure T_i directly from the dates. Otherwise, I predict exclusivity using the protection most likely to block entry, typically the active ingredient's primary patent. If no primary patent is identifiable, I use the extended patent as a proxy. It provides the active ingredient's primary protection in 79 percent of cases (Hemphill and Sampat 2012). I account for development and review time, then add the

Figure 4: Key moments in market exclusivity



Notes: Effective market exclusivity, T_i, runs from NDA approval to generic entry. FDA review time, R_i, runs from NDA submission to NDA approval and is the instrument of Section 5.4. The patent term extension restores part of the regulatory review period, up to five years.

adjusted PTE. The predicted measure is

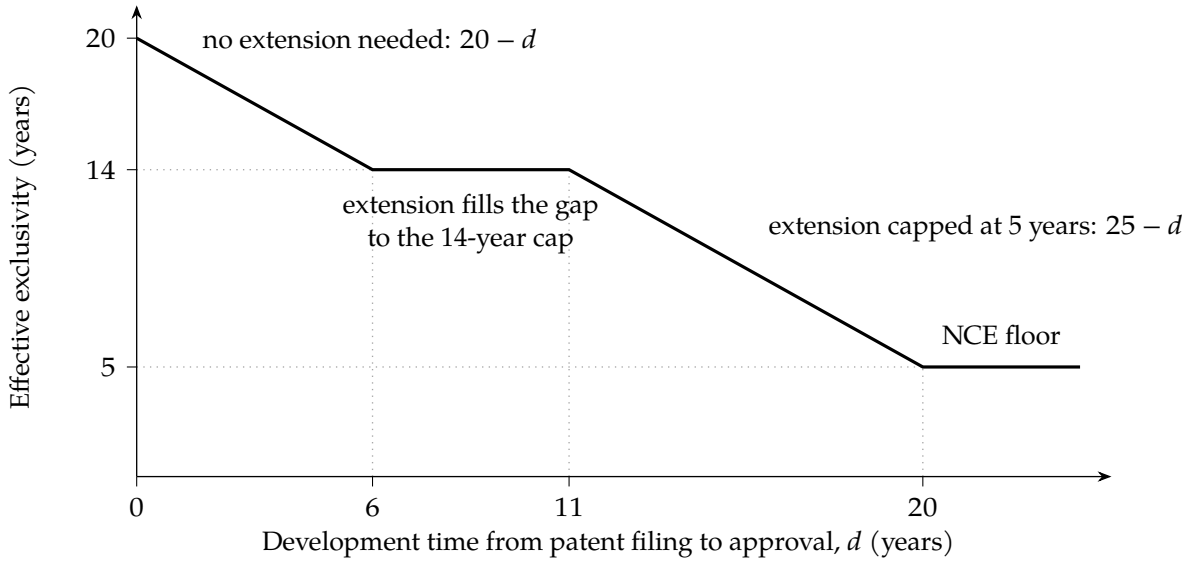
$$\widehat{T}_i \;=\; (20 \text{ years} - \text{development time}_i) + \text{adjusted PTE}_i \tag{9}$$

where development time is the interval from the associated patent's priority date to the drug's first approval, and adjusted PTE restores up to five of the years lost to development, subject to the statutory cap of fourteen years of effective post-approval patent life. Longer development mechanically consumes patent life, which makes PTE the binding protection for slow-developing drugs (Figure 5).

4.3 The Start of Clinical Development

I date clinical development from each NME's Investigational New Drug (IND) application approval. Development timelines are usually proprietary, but PTE applicants must certify the regulatory review period's start and end dates. The USPTO patent-term-extension record publishes this certification, including the IND approval date. I retrieve each extension record by patent number and convert the documents to machine-readable text. Document-specific keyword searches identify the drug name, patent number, and IND approval date. This source provides dates only for molecules whose sponsors sought and

Figure 5: Effective market exclusivity as a function of development time



Notes: Stylized relation implied by the statute. Without an extension, patent life after approval is $20 - d$. The extension restores up to five years but cannot raise post-approval patent life above fourteen years, so it is zero for $d \leq 6$, fills the gap to the cap for $6 \leq d \leq 11$, and is capped at five years thereafter. New chemical entity exclusivity sets a floor of five years.

obtained an extension. Section 7 quantifies this restriction.

4.4 Disease Classification

Each NME's initial indication is mapped from free text to four-digit International Classification of Diseases (ICD) codes. First, I standardize the descriptions by merging the NME population with the sample in Gilchrist (2016), which contains standardized initial indications. Second, I map these descriptions to ICD codes. I use FDA approval data to identify initial orphan indications.

4.5 Matching Firms across Events

Sponsors are classified as the same firm after accounting for parent–subsidiary relationships, mergers, and acquisitions. Sponsor names are recorded at first approval for each NME and at indication approval for each repurposing event. These names are resolved in four steps. First, I standardize corporate suffixes and abbreviations. Second, I use Claude Opus 4.5 with web search to identify mergers and acquisitions and their years. I separate acquired

firms from those never acquired through 2024. Third, I use the same procedure to identify ultimate parents for firms never acquired. Fourth, I replace recorded names with the ultimate parent or the acquirer as of the event year. An event is classified as originator if the resolved names match and as entrant otherwise. Every model-assisted assignment is cross-checked independently by Gemini Flash, and I manually audit repeated random samples of ten NMEs.

4.6 Sample and Summary Statistics

The population contains 1,342 NMEs approved between 1985 and 2024, each carrying an active moiety approved by the FDA for the first time. Four restrictions define the estimation sample.

I retain 1,075 small molecules and exclude 267 biologics, following the institutional mapping in Section 3.2. The United States had no statutory biosimilar pathway until the Biologics Price Competition and Innovation Act (BPCIA), enacted on 23 March 2010. Biologics therefore span two regimes, neither providing an observable end of protection. For the 1985–2010 cohort, competitors had no abbreviated route relying on originator data, so there is no dated expiry to measure. For the post-2010 cohort, the BPCIA provides twelve years of reference-product exclusivity. Biologics approved late in the window have typically not completed this period. Protection length is right-censored in both cases.

The Orange Book also covers only small-molecule products. Biologic rights appear in the Purple Book, which lacks the patent and exclusivity detail required for my measure. Excluding biologics therefore restricts scope rather than creates sample-selection concerns: the estimates apply to the small-molecule regime.

I then exclude 68 fixed-dose combinations and 18 molecules subsequently withdrawn from the market, leaving 989 molecules. For combinations, protection relevant to a follow-on use cannot be attributed to one active moiety. Withdrawn molecules have shorter commercial lives, making their post-approval indication counts non-comparable.

I exclude another 294 molecules whose protection end dates cannot be measured. They have neither observed generic entry nor expected exclusivity dated by an associated extended patent's expiry. The estimation sample therefore contains 695 drugs.

Table 3 reports summary statistics for this common sample. Mean post-approval indications are 1.41 per molecule, and mean effective exclusivity is 12.54 years.

Table 3: Summary Statistics

	N	Mean	SD	Min	Max
Repurposing events (count)	695	1.410	2.469	0.00	31.00
Any repurposing (1/0)	695	0.496	0.500	0.00	1.00
Market exclusivity (years)	695	12.540	4.612	5.31	39.88
Patents at approval	695	4.016	4.412	0.00	35.00
Years on market	695	22.089	10.121	3.00	39.00
Orphan drug (1/0)	695	0.212	0.409	0.00	1.00
Priority review (1/0)	695	0.426	0.495	0.00	1.00
FDA review time (years)	695	1.739	1.617	0.20	15.30
Patent priority to IND (years)	220	5.545	3.637	-5.72	20.38

Notes: Estimation sample: 695 small-molecule new molecular entities approved by the FDA in 1985–2024. Repurposing events are post-approval new-indication exclusivities in the Orange Book. The priority-to-IND interval needs an IND date and is observed for 220 drugs.

5 Empirical Strategy

5.1 From Propositions to Testable Hypotheses

Section 2.2 provides the null hypothesis: the upstream patent acts only as a gate that obstructs new uses. The alternative in Section 2.3 adds protection for each follow-on use. Propositions 1–3 distinguish their predictions for the effect of longer upstream protection on development counts (Proposition 1), across follow-on products that carry different protection of their own (Proposition 2), and across developers (Proposition 3). Table 4 summarizes the predictions.

Hypothesis 1a (level effect; Corollary 1 and Proposition 1). Galasso and Schankerman (2015) find that patent invalidation raises follow-on innovation in complex technology fields but not in pharmaceuticals. This suggests low pharmaceutical licensing frictions and a large ρ. Proposition 1 then predicts that longer market exclusivity raises the number of follow-on indications: $\beta > 0$ in the instrumented count regression of Sections 5.2 and 5.4, whereas the gate-only null predicts $\beta \leq 0$.

Hypothesis 1b (shape; Corollary 1 and Proposition 1). Proposition 1 links the response profile to the opportunity elasticity: concave if $\eta < 1$, linear if $\eta = 1$, and convex if $\eta > 1$. The gate-only null predicts a linear response.

Hypothesis 2a (product form; Proposition 2). A new indication reaches the market in one of three product forms (Table 1), which differ in how much protection the new use carries

on its own. A new indication added to an existing product is the same pill a generic sells, so generics serve it directly through substitution and off-label prescribing: its h is the lowest. A new dosage form is protected by formulation patents and is not automatically substituted: its h is intermediate. A combination adds a second active ingredient that single-molecule generics cannot replicate: its h is the highest. Proposition 2 therefore predicts that the response to T_1 is largest for same-product indications, smaller for new dosage forms, and smallest for combinations. Under the gate-only null, the follow-on's own protection does not matter and the three responses are equal.

Hypothesis 2b (rare and non-rare indications; Proposition 2). Rare-disease indications receive a secondary patent and seven years of orphan drug exclusivity; other indications receive a secondary patent and three years of clinical-investigation exclusivity. Orphan exclusivity bars the FDA from approving another sponsor's application for the same drug and orphan indication; clinical-investigation exclusivity bars it from relying on the innovator's new clinical data. Each protects only the indication, not the drug, so neither reaches off-label or cross-label generic use. If this indication-level protection is weak in practice, the response to T_1 is positive and similar for rare and non-rare indications. If it binds, orphan indications have the higher h, and Proposition 2 predicts a smaller response for them. The gate-only null predicts no positive response in either market.

Hypothesis 3 (originators and entrants; Proposition 3). When licensing frictions are small, originators and entrants both respond positively to T_1. Originators benefit from protection without needing a license. Entrants respond positively because protection outweighs licensing barriers when ρ is high. The gate-only null predicts no response from originators and a negative or zero response from entrants.

Table 4: Model predictions and the gate-only null

	Object tested	**Asymmetric model** (H_1)	**Gate-only null** (H_0)
H1a	Average effect	$\partial n/\partial T_1 > 0$	≤ 0
H1b	Shape of the marginal response	Concave if $\eta < 1$; linear if $\eta = 1$; convex if $\eta > 1$	Linear
H2a	Product form	Smaller response where h is higher	No difference
H2b	Rare vs. non-rare indications	Smaller response where h is higher	No difference

	Object tested	Asymmetric model (H_1)	Gate-only null (H_0)
H3a	Originator response	> 0	$= 0$
H3b	Entrant response	> 0 when ρ is high	≤ 0

5.2 Baseline Specification

The paper examines how the duration of market exclusivity of the first-approved drug affects subsequent repurposing. With i indexing drugs containing a first-approved active ingredient, the baseline model is the exponential conditional mean

$$\mathbb{E}[\, N_i \mid T_i, X_i \,] \;=\; \exp\!\big(\alpha + \beta\, T_i + X_i'\xi\big)\,, \tag{10}$$

estimated by Poisson quasi-maximum-likelihood (QML). Here N_i counts drug i's post-approval indications (Section 4.1), and T_i measures its effective market exclusivity in years (Section 4.2). The control vector X_i (Section 5.3) includes the drug's patent count before approval and years on the market through 2024. The coefficient β is a semi-elasticity. An additional year of exclusivity changes expected repurposing by $(e^{\beta} - 1) \cdot 100$ percent, approximately $100\,\beta$ percent for small β.

5.3 Control Variables

There are three threats to causal inference. First, confounding may bias the estimate in either direction. High-value drugs might be associated with more potential new uses, and firms may extend their exclusivity through evergreening (Hemphill and Sampat 2012). Meanwhile, those high-value drugs may attract more Paragraph IV challenges by generic firms, which shorten market exclusivity. Second, measurement error understates the effect: for drugs without observed generic entry, I measure effective exclusivity using the extended patent. Third, anticipation also understates the effect. In-silico screening in the 1990s and large electronic medical records in the 2010s have reduced repurposing costs. Firms in recent cohorts may shift development from post-approval repurposing to pre-approval repositioning.

The first control variable, for the first threat, is the number of patents associated with the drug before its first approval. Because patents filed after approval include patents on the

new uses themselves, controlling for the full patent stock would partly control for the outcome; I therefore use the number of patents associated with the drug before its first approval.

The second is time on the market. A drug marketed longer has had more time to accumulate new indications, and approval cohorts differ in the macro environment they faced, for instance in the technologies available to them; I control for the drug's years on the market through 2024.

The third is the therapeutic area of the first approval. Trial durations and success rates differ sharply across disease areas, oncology trials for instance being longer and more failure-prone than cardiovascular or infectious-disease trials, and some areas generate spillovers to others, as when platform technologies developed for rare diseases seed non-orphan applications. I control for the ICD area of the first indication and for the drug's orphan status at first approval.

5.4 Instrumental Variable

I instrument effective market exclusivity with FDA review time, R_i, the interval between NDA submission and approval. Section 7 uses the priority-to-IND gap of Gilchrist (2016) as an alternative instrument.

Review time shortens exclusivity only where patent term restoration fails. Under 35 U.S.C. §156 the extension credits half of the testing phase and all of the review period, subject to a five-year cap on the extension and a fourteen-year cap on exclusivity remaining after approval:

$$T_i = \min\{\, t_i^{\text{pat}} + \text{PTE}_i,\ t_i^{\text{gen}},\ t_i^{\text{app}} + 14 \,\} - t_i^{\text{app}}, \qquad \text{PTE}_i = \min\{\tfrac{1}{2}\,\text{test}_i + R_i,\ 5\},$$

where $t_i^{\text{app}} = t_i^{\text{sub}} + R_i$ is the approval date, t_i^{sub} the submission date, t_i^{pat} nominal patent expiry, test_i the length of the testing phase and t_i^{gen} the date of generic entry. When restored patent expiry binds and the cap does not, R_i cancels and $\partial T_i / \partial R_i = 0$. When exclusivity ends at generic entry or at the five-year cap, its end date is fixed and $\partial T_i / \partial R_i = -1$. The first stage is therefore driven by the second regime. The first-stage coefficient is $\hat{\omega} = -0.543$ (s.e. 0.099; partial $F = 30.07$; Appendix Table B.1): one more year of review reduces exclusivity by about half a year. As the arithmetic predicts, the coefficient is strongest among molecules whose exclusivity ends at observed generic entry (-0.61) and zero among molecules held at the fourteen-year ceiling.

I rely on review-time variation that reflects FDA capacity rather than application characteristics. Before 1992, median NDA review took about 29 months, with the backlog attributed to insufficient staff. The Prescription Drug User Fee Act (PDUFA) I (1992) allowed the FDA to collect user fees to hire reviewers without changing approval standards. In exchange, it set review goals of 12 months for standard applications and 6 months for priority applications. FDAMA/PDUFA II (1997) phased the standard goal down to 10 months. FDASIA/PDUFA V (2012) started the clock for new molecular entities after a 60-day filing period, making the effective goals 12 and 8 months. User fees reduced review times beyond the pre-existing trend. Berndt et al. (2005) estimate additional annual reductions of 6–7 percent under PDUFA I and 3–4 percent under PDUFA II. Darrow, Avorn, and Kesselheim (2017) also document faster review following user fees. The controls remove a linear approval-cohort trend. Because X_i includes years on market, a linear function of approval year, review-time deviations from that trend identify β. About 84 percent of the variance in R_i is within approval year. Appendix Figure B.1 plots review time by approval year: the median falls after PDUFA. To account for approval-year differences beyond a linear trend, I replace this control with approval-year fixed effects. Identification then uses only review-time differences among drugs approved in the same year, yielding $\hat{\beta} = 0.314$ (s.e. 0.095).

Clinical importance is the main observable threat. Priority review is associated with shorter review conditional on other controls ($t = -5.4$) and more repurposing events (1.59 versus 1.28). If review speed proxies for clinical importance, $\hat{\beta}$ is biased upward. I include priority review in X_i to address this threat. The estimate changes little, from 0.294 without this control to 0.291 with it. The effect remains among standard-review drugs (0.245, s.e. 0.103). Conditional on priority review, orphan status is unrelated to review time ($t = -0.7$).

The instrumental-variable estimate is local to molecules whose patent term extension does not fully restore protection lost during review. These reach the five-year extension cap or lose exclusivity at generic entry. The estimate need not apply when lost review time is fully restored, including at the fourteen-year ceiling, where review time does not affect exclusivity.

The first stage regresses T_i on R_i and the controls using the estimation sample in Section 4.6. I report two-stage least squares for the linear model. For the count model, I use a control function for the exponential conditional mean, adding the first-stage residual $\hat{v}_i$ to (10).

5.5 Stacked Specifications for the Heterogeneity Hypotheses

Hypotheses 2 and 3 compare exclusivity semi-elasticities across product forms, across rare and non-rare indications, and across originator and entrant events. For each comparison, I stack two rows per drug, one per group k. Let N_{ik} denote drug i's repurposing count in group k. The stacked conditional mean is

$$\mathbb{E}[\, N_{ik} \mid \cdot \,] \;=\; \exp(\alpha_k + \beta_k\, T_i + X_i'\xi_k + \lambda_k\, \hat{v}_i), \tag{11}$$

where regressors vary only by molecule and all coefficients are group-specific, making the specification fully interacted. β_k is group k's semi-elasticity. The group difference is $\delta \equiv \beta_{\text{nonrare}} - \beta_{\text{rare}}$ for the disease split and $\delta \equiv \beta_O - \beta_E$ for the filer split. It is the coefficient on T_i interacted with the indicator for the first group in the difference. Its standard error is used to test equality. For product forms, I estimate two such models, stacking new-dosage-form and combination events in turn against same-product events, so that $\delta_k \equiv \beta_k - \beta_{\text{same}}$. The control function $\hat{v}_i$ is the first-stage residual from Section 5.4.

6 Results

6.1 The Level Effect of Exclusivity (H1a)

Table 5 estimates the average effect of market exclusivity on follow-on repurposing. It has four columns: OLS, Poisson QML from equation (10), linear two-stage least squares, and a control-function IV-Poisson. The control-function IV-Poisson is estimated in two steps:

$$\begin{aligned} T_i &= \omega\, R_i + X_i'\zeta + v_i, \\ \mathbb{E}[N_i \mid T_i, X_i, \hat{v}_i] &= \exp\!\left(\alpha + \beta\, T_i + X_i'\xi + \lambda\, \hat{v}_i\right). \end{aligned}$$

The first stage regresses exclusivity on FDA review time and the controls, and the second stage adds the first-stage residual $\hat{v}_i$ to the Poisson model to absorb exclusivity's correlation with unobserved factors, so that only review-time-driven variation in exclusivity identifies β. The t-statistic on λ tests whether exclusivity is exogenous.

The uninstrumented estimates are small and insignificant: 0.007 (0.020) for OLS and 0.006 (0.013) for Poisson. Instrumenting raises both. The linear 2SLS coefficient is 0.277 (0.081), equivalent to 0.277 additional indications per year of exclusivity, about 20 percent of the

Table 5: Effect of Market Exclusivity on Repurposing

	Dependent variable: repurposing events (count)			
	OLS	Poisson	IV-2SLS	IV-Poisson
Market exclusivity (years)	0.007	0.006	0.277***	0.291***
	(0.020)	(0.013)	(0.081)	(0.106)
Years on market	0.043***	0.030***	-0.003	-0.018
	(0.009)	(0.006)	(0.017)	(0.020)
Patents at approval	0.099***	0.057***	0.081***	0.040**
	(0.026)	(0.010)	(0.029)	(0.017)
Orphan drug	-0.129	-0.106	-0.363	-0.348
	(0.190)	(0.139)	(0.239)	(0.215)
Priority review	0.294	0.196	0.181	0.071
	(0.195)	(0.135)	(0.222)	(0.186)
First-stage F			30.07	30.07
Control-function t				-2.77
(Pseudo) R-squared	0.039	0.037		
Observations	695	695	695	695

Notes: Dependent variable: post-approval repurposing events per drug. Columns 3 and 4 instrument exclusivity with FDA review time; column 4 is a two-step control-function Poisson. All columns control for years on market, patents at approval, orphan status and priority review. Robust standard errors in columns 1 to 3; in column 4, bootstrap standard errors over both stages (999 replications). The control-function t tests whether exclusivity is exogenous. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

sample mean of 1.41, and the IV-Poisson semi-elasticity is 0.291 (0.106), my preferred estimate because the outcome is a count. The increase is consistent with the downward biases from anticipation and measurement error described in Section 5.3, and the control-function t-statistic of -2.77 rejects exogeneity, indicating that the uninstrumented estimates are biased. Both instrumented estimates are positive, whereas the gate-only view predicts $\beta \leq 0$.

An additional year of effective exclusivity raises expected post-approval indications by $(e^{0.291} - 1) \cdot 100 \approx 34$ percent (column 4). This corresponds to about 0.48 additional indications at the sample mean of 1.41 events per molecule. The positive effect rejects the gate-only null and is consistent with upstream patent protection against imitation.

The first-stage coefficient on review time is -0.543 (0.099; Appendix Table B.1): an additional review year reduces exclusivity by about half a year. As Section 5.4 explains, the estimate is local to drugs whose patent term extension does not fully restore protection lost

during review.

6.2 The Shape of the Effect (H1b)

I allow the response to vary with the length of exclusivity in two control-function Poisson models, $\mathbb{E}[N_i \mid \cdot] = \exp\left(g(T_i) + X_i'\xi + \text{CF terms}\right)$, with

$$\begin{aligned} \text{quadratic:} \quad & g(T) = \beta_1 T + \beta_2 T^2, \\ \text{spline:} \quad & g(T) = b_1 \min(T, 11) + b_2 \max\{\min(T, 16) - 11,\, 0\} + b_3 \max(T - 16,\, 0). \end{aligned}$$

The spline is piecewise linear with knots at 11 and 16 years, so b_1, b_2 and b_3 are the slopes below 11 years, between 11 and 16 years, and above 16 years. Nonlinear entry of T_i requires more than a single residual to control for endogeneity. I include the first-stage residual $\hat{v}_i$ from the regression on FDA review time, its square, and its interaction with T_i:

$$\text{CF terms} = \lambda_1 \hat{v}_i + \lambda_2 \hat{v}_i^2 + \lambda_3 \hat{v}_i T_i$$

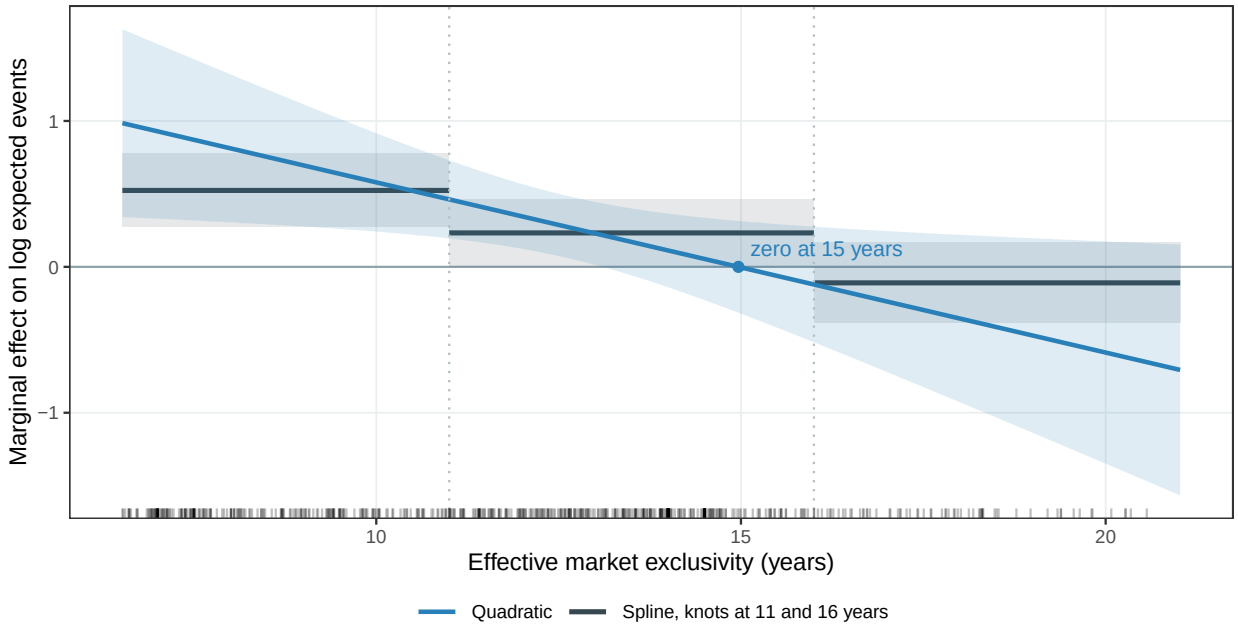
Figure 6 plots the marginal effect each model implies; Appendix Table B.2 reports the coefficients.

The quadratic estimates imply $\eta < 1$ and a declining marginal effect. In the model, β_2 is negative when $\eta < 1$ (decreasing returns), zero when $\eta = 1$ (constant returns), and positive when $\eta > 1$ (increasing returns). The quadratic gives $\beta_1 = 1.746$ (0.649) and $\beta_2 = -0.058$ (0.025), with the latter significant at the 5 percent level. The negative β_2 implies increasing marginal development costs, with lower-cost uses developed first. The marginal effect $\beta_1 + 2\beta_2 T$ declines with exclusivity and reaches zero at $T^* = -\beta_1/(2\beta_2) \approx 15$ years.

The spline estimates show a similar decline. Below 11 years, the slope is 0.524 (0.129), equivalent to $(e^{0.524} - 1) \cdot 100 \approx 69$ percent more events per additional year. Between 11 and 16 years, it falls to 0.233 (0.119), or about 26 percent, significant at the 5 percent level. Above 16 years, the slope is -0.110 (0.140) and not significant. The quadratic's turning point, about 15 years, lies where the spline moves from a positive slope to one indistinguishable from zero.

Longer exclusivity protects returns to new uses but delays entry for developers who must license. At short durations, the marginal protection benefit dominates, producing a large positive net effect. With decreasing returns ($\eta < 1$), each additional year induces fewer new uses. At long durations, the marginal protection benefit becomes small enough for

Figure 6: Marginal effect of exclusivity, by its length



Notes: Marginal effect of one more year of exclusivity on log expected repurposing events (a semi-elasticity), from control-function Poisson regressions with exclusivity instrumented by FDA review time. The line is $\beta_1 + 2\beta_2 T$ from the quadratic, with its 95 percent confidence band; the horizontal segments are the three spline slopes, with 95 percent confidence intervals shaded. Confidence intervals come from a bootstrap over both stages. The curve spans the 5th to 95th percentiles of exclusivity; ticks on the horizontal axis mark individual drugs.

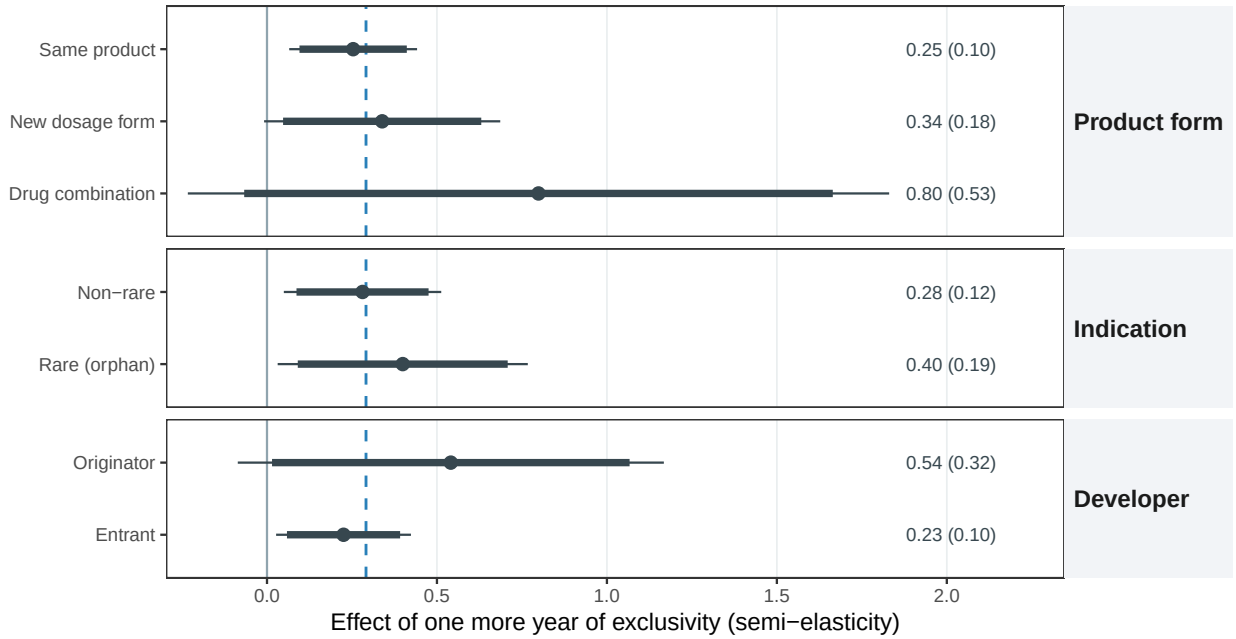
licensing delays to offset it. The insignificant slope above 16 years is consistent with the two effects roughly offsetting each other. It does not establish that licensing delays dominate.

6.3 Heterogeneous Responses (H2 and H3)

Figure 7 summarizes the three heterogeneity tests, each estimated with the stacked specification (11); Appendix Tables B.3–B.5 report the full estimates. The response to exclusivity is positive in every group.

Product form (H2a). For same-product repurposing, where the new use carries the least protection of its own, the coefficient is 0.253 (0.096), or $(e^{0.253} - 1) \cdot 100 \approx 29$ percent more events per additional year of exclusivity, significant at the 1 percent level. For new dosage forms, the coefficient is 0.339 (0.177), an increase of roughly 40 percent, significant only at the 10 percent level. For drug combinations, which carry the most protection of their own, the coefficient of 0.799 (0.526) is statistically insignificant. The estimated differences from same-product repurposing, $\delta_k = \beta_k - \beta_{\text{same}}$, are positive: 0.085 (0.153) for new dosage

Figure 7: Exclusivity response by product form, indication and developer



Notes: Each point is the effect of one more year of market exclusivity on log expected repurposing events in one group, from the stacked control-function Poisson regressions of Section 5.5, with exclusivity instrumented by FDA review time. Thick bars are 90 percent and thin bars 95 percent confidence intervals, from a drug-level bootstrap over both stages; the numbers on the right are the estimate and its standard error. The dashed line is the pooled estimate (Table 5, column 4).

forms and 0.545 (0.513) for drug combinations. Proposition 2 predicts negative differences, but neither is statistically significant, so the data cannot reject the predicted ordering. Because the two stacks share the same-product arm, these differences are not independent and cannot be pooled for a joint test. Statistical significance declines as own protection rises, which fits Proposition 2, but the large standard errors preclude ruling out sizable differences in either direction.

Rare and non-rare indications (H2b). The effect is positive in both markets. For rare indications, the coefficient is 0.399 (0.188), or $(e^{0.399} - 1) \cdot 100 \approx 49$ percent more events per additional year of exclusivity, significant at the 5 percent level. For non-rare indications, the coefficient is 0.281 (0.118), or about 32 percent, also significant at the 5 percent level. The difference between the two, $\delta = \beta_{\text{nonrare}} - \beta_{\text{rare}}$, is −0.118 (0.197) and not significant, so I find no evidence that the effect differs between rare and non-rare markets, although the test cannot rule out a sizable difference either.

One might have expected a weaker response for rare indications, because seven-year orphan drug exclusivity (ODE) could reduce their dependence on molecule-level exclusivity. Yet ODE blocks only FDA approval of the same drug for the same indication. Once

generics enter, physicians can prescribe them off-label for the orphan indication. ODE therefore provides little practical protection, leaving rare-disease repurposing dependent on molecule-level exclusivity. The result suggests that FDA indication-level exclusivities add little incentive to develop new uses for approved drugs. This interpretation remains tentative because the estimated difference is insignificant.

Originators and entrants (H3). For originator events the coefficient is 0.541 (0.320), or $(e^{0.541} - 1) \cdot 100 \approx 72$ percent more originator events per additional year of exclusivity, significant at the 10 percent level. The gate-only null is rejected at that level: a firm's repurposing of its own molecule is not invariant to how long that molecule's protection runs. The entrant coefficient is 0.225 (0.101), significant at the 5 percent level, implying $(e^{0.225} - 1) \cdot 100 \approx 25$ percent more entrant events per year of exclusivity. This positive response contradicts the gate-only prediction of a strictly negative one, under which longer upstream protection only imposes licensing costs and delays; it is consistent with the shield model, which allows a positive entrant response when licensing transaction costs are low. The originator coefficient is larger, as Proposition 3 predicts, but the difference, 0.316 (0.327), is not statistically significant.

The control estimates differ across originator and entrant events (Appendix Table B.5). Years on market has a negative coefficient for originators (-0.107, significant at the 10 percent level) and a small, insignificant coefficient for entrants (0.024). Because counts are cumulative, the negative coefficient reflects fewer originator events in earlier approval cohorts, rather than declining counts as individual drugs age. Patent count at approval matters only for entrant events (0.054, s.e. 0.023). Priority review is insignificant for both.

7 Robustness: An Alternative Instrument

Following Gilchrist (2016), I instrument market exclusivity with the *priority-to-IND gap*: the interval between the priority date of the drug's extended patent (Section 4.2) and the start of its clinical trials, dated by the IND approval of Section 4.3.

Patent term extension restores part of clinical-trial time and all FDA review time, but none of the interval between patent filing and IND. A longer priority-to-IND gap therefore shortens effective patent life and market exclusivity, as commercialization lags do (Budish, Roin, and Williams 2015). This instrument uses a different pre-market interval from review time and faces different threats to the exclusion restriction. Similar estimates would suggest that a review-time-specific problem does not drive the main result.

The exclusion restriction requires the priority-to-IND gap to be unrelated to later repurposing potential, conditional on controls. This is plausible for three reasons. First, drug discovery is non-linear. Molecules developed for one purpose often prove effective for another through unexpected discoveries during or after clinical trials. Early filings reflect uncertainty about eventual uses rather than repurposing plans. Second, competition discourages delayed filing. Firms file when a molecule shows promise, regardless of its commercial prospects, separating filing timing from exclusivity strategy. Third, pressure to publish encourages early filing because firms must file before public disclosure.

The gap requires an IND date, which I observe for only 220 of the 695 molecules. Table 6 has three columns. Column 1 is the main specification, instrumented by review time on the full sample. Column 2 uses the priority-to-IND gap as the instrument on the 220 molecules with an IND date. Column 3 uses both instruments on the same 220 molecules; Appendix Table B.6 reports the first stages.

Table 6: The Priority-to-IND Gap as an Alternative Instrument

	All molecules	Molecules with an IND date	
	Review time	Priority to IND	Both
Market exclusivity (years)	0.291***	0.270**	0.175**
	(0.106)	(0.128)	(0.083)
Years on market	-0.018	0.008	0.019
	(0.020)	(0.018)	(0.017)
Patents at approval	0.040**	0.107**	0.089**
	(0.017)	(0.046)	(0.038)
Orphan drug	-0.348	-0.195	-0.171
	(0.215)	(0.351)	(0.313)
Priority review	0.071	0.188	0.217
	(0.186)	(0.299)	(0.276)
First-stage F	30.07	24.82	20.26
Control-function t	-2.77	-2.08	-2.00
Over-identification p-value			0.187
Observations	695	220	220

Notes: Dependent variable: post-approval repurposing events per drug. Control-function Poisson; the column heads name the excluded instrument, and columns 2 and 3 are restricted to molecules with an IND date. The over-identification p-value is from the Hansen J test of the corresponding 2SLS. Controls as in Table 5. Bootstrap standard errors over both stages (999 replications). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Exclusivity has a positive effect, significant at the 5 percent level, in columns 2 and 3.

Column 2 gives a coefficient of 0.270, close to the main estimate of 0.291. This implies about 31 percent more repurposing events per additional year of exclusivity. The first-stage F statistic is 24.82, indicating a strong instrument. Column 3 gives a smaller coefficient of 0.175 and a first-stage F statistic of 20.26. The Hansen J statistic is 1.74, with a p-value of 0.19. The instruments shift exclusivity for different molecules: the gap where statutory caps do not bind, and review time where patent term restoration is incomplete. The test therefore compares the two groups' identified effects rather than validating either instrument separately. Its non-rejection suggests similar local effects. The comparison between columns 1 and 2 changes both the instrument and the sample, since column 2 uses fewer observations. The similar estimates nevertheless suggest that the main result does not depend on instrument choice.

Appendix C varies how the regressions account for the observation window: an exposure term, a quadratic in years on market, and a sample restricted to drugs observed for at least ten years. The estimate stays between 0.226 and 0.291 and significant at the 5 percent level in every specification except the pure exposure model, whose proportionality assumption the data reject.

8 Conclusion

Intellectual property in pharmaceuticals is asymmetric across generations of innovation. A molecule's patent covers every use, while the right on a new use is narrow and hard to enforce once generics enter. This paper shows that the asymmetry changes what the upstream patent does. Besides requiring follow-on innovators to obtain a license, it provides their main protection against imitation, because imitating a licensed follow-on uses the molecule and infringes the upstream patent. The gate view of cumulative innovation is the special case of symmetric protection, in which the follow-on's own right excludes imitation and longer upstream protection can only delay follow-on innovation.

The evidence indicates that the asymmetry binds. Instrumenting effective exclusivity with FDA review time, I find that an additional year raises expected repurposing by about 34 percent, and the priority-to-IND gap yields a similar estimate. The marginal effect is largest at short durations and indistinguishable from zero beyond about 15 years. Originators, who need no license and would not respond under symmetric protection, respond positively, as do entrants. The effect is precisely estimated for new indications added to the same product, which generics substitute directly; reformulations and combinations, which carry more

protection of their own, show positive but imprecise responses that cannot confirm the predicted ordering. Rare-disease indications respond similarly to others despite seven-year orphan exclusivity. The pattern in Figure 1 fits this reading: new uses accumulate while the molecule is protected, and nothing replaces that protection when it lapses.

The policy problem is the asymmetry, not patent length. Upstream exclusivity stimulates repurposing only because the right on a new use cannot exclude imitation on its own. Extending upstream protection would add little, since the marginal effect is close to zero beyond about 15 years. Lengthening indication-level exclusivity is also unlikely to close the gap, because such exclusivity does not reach off-label generic use. Remedies should target the asymmetry, by making the return to a new use enforceable, for example through differentiated prices for the new use, or by replacing that return with public funding for repurposing trials.

Three limitations qualify these results. First, the Orange Book lists only small-molecule products, so the analysis excludes biologics, whose protection follows a separate regime and is not observed in a comparable form. Second, the analysis captures only new uses that receive FDA approval and excludes trials that non-profit organizations and universities may conduct after generic entry; the scarcity of subsequent approvals nevertheless reflects weak commercial incentives to bring new uses to the label. Third, the instrumental-variable estimate is local to drugs whose patent term extension does not fully restore lost review time and may not apply where it does.

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Acknowledgement of AI Use

I used large language models for three tasks: (i) Language. Claude was used to check grammar and to suggest clearer wording for sentences that I had drafted; I reviewed each suggestion and kept, changed, or discarded it. (ii) Data construction. As described in Section 4.5, Claude Opus 4.5 with web search was used to look up merger, acquisition, and parent–subsidiary relationships for the sponsor names in the data, and Gemini Flash (Google) independently cross-checked every assignment; I audited repeated random

samples of the assignments by hand. (iii) Model. Claude was used to check the algebra of the lemmas and propositions, the consistency of notation across sections, and whether each simplifying assumption is stated where it is used; I re-derived every result that appears in the paper. No passage, result, or argument was included without my review, and I take full responsibility for the content.

A Proofs

Proof of Lemma 1. (i) In the shielded phase the incumbent's patent excludes the fringe, so a consumer's outside option is not buying, worth zero. A consumer of type θ buys at price p if and only if $v + \theta\Delta - p \geq 0$, that is, $\theta \geq (p - v)/\Delta$. Demand is therefore $e^{-(p-v)/(m\Delta)}$ for $p \geq v$. The developer maximises $p\, e^{-(p-v)/(m\Delta)}$. The first-order condition gives $p_S = m\Delta$, which is interior since $m\Delta > v$ by (A1). Coverage is $e^{-(m\Delta - v)/(m\Delta)} = e^{z-1}$ and $\pi_S = m\Delta\, e^{z-1}$.

(ii) In the exposed phase the best legal imitation, of quality $\Delta - h$ at price zero, is every consumer's outside option, worth $v + \theta(\Delta - h)$. A consumer buys from the developer if and only if $v + \theta\Delta - p \geq v + \theta(\Delta - h)$, that is, $\theta \geq p/h$. The baseline v and the unenclosed increment $\theta(\Delta - h)$ cancel. Demand is $e^{-p/(mh)}$. Maximising $p\, e^{-p/(mh)}$ gives $p_X = mh$, share e^{-1}, and $\pi_X = mh\, e^{-1}$. Every consumer with $\theta < p_X/h$ buys the free imitation, so the market is fully covered. □

Proof of Lemma 2. By Lemma 1, $\phi = \pi_X/\pi_S = (mh\, e^{-1})/(m\Delta\, e^{z-1}) = (h/\Delta)\, e^{-z} = \kappa\, e^{-z}$. Under (A4) $\kappa \leq 1$, and under (A1) $z \in (0, 1)$, so $\phi \leq e^{-z} < 1$. □

Proof of Corollary 1. In (5) only the factor $e^{-\eta r T_1}$ depends on T_1, and its derivative is $-\eta r\, e^{-\eta r T_1}$. Multiplying by $\gamma_i (\pi^* A)^\eta (1 - \rho)$ gives the displayed derivative. Every factor is positive except $-(1 - \rho)$, so the derivative is strictly negative for $\rho < 1$ and identically zero at $\rho = 1$. □

Proof of Proposition 1. Step 1: a threshold in ρ. By (8), $\partial n/\partial T_1$ has the sign of B. Write $B = \rho\, a - (1 - \rho)\, c$ with

$$a \;\equiv\; (\pi_S - \pi_X)\, \Pi_{\text{succ}}^{\eta-1} \;>\; 0, \qquad c \;\equiv\; rK\, \Pi_{\text{fail}}^{\eta-1} \;>\; 0,$$

where $a > 0$ uses $\pi_S > \pi_X$ from Lemma 2. Then $B > 0$ if and only if $\rho\,(a + c) > c$, that is, $\rho > c/(a + c) \equiv \hat{\rho}(T_1)$, which lies in $(0, 1)$.

Step 2: the threshold at expiry. At $T_1 = 0$ the two branches hold the same window: $\Pi_{\text{succ}}(0) = \pi_X D(0, T_2) = K = \Pi_{\text{fail}}(0)$. So $a = (\pi_S - \pi_X) K^{\eta-1}$ and $c = rK \cdot K^{\eta-1}$. The common factor cancels in the ratio, and $\hat{\rho}(0) = rK/(rK + \pi_S - \pi_X)$, which does not involve η.

Step 3: the threshold as T_1 grows. Using $\partial \Pi_{\text{succ}}/\partial T_1 = (\pi_S - \pi_X) e^{-rT_1}$ and $\partial \Pi_{\text{fail}}/\partial T_1 = -rK e^{-rT_1}$,

$$\frac{\partial a}{\partial T_1} \;=\; (\eta - 1)\, (\pi_S - \pi_X)^2\, e^{-rT_1}\, \Pi_{\text{succ}}^{\eta-2}, \qquad \frac{\partial c}{\partial T_1} \;=\; -\,(\eta - 1)\, r^2 K^2\, e^{-rT_1}\, \Pi_{\text{fail}}^{\eta-2}.$$

So a and c move in opposite directions. If $\eta < 1$, a falls and c rises. If $\eta > 1$, a rises and c falls. If $\eta = 1$, both are constant. Since $\hat{\rho} = c/(a+c) = 1/(1 + a/c)$ decreases in a/c, $\hat{\rho}$ is strictly increasing in T_1 when $\eta < 1$, strictly decreasing when $\eta > 1$, and constant when $\eta = 1$.

Step 4: the three cases. (i) If $\eta = 1$, then $\hat{\rho} \equiv \rho_{\text{lo}}$, so B has the sign of $\rho - \rho_{\text{lo}}$ at every T_1 and n is monotone.

(ii) If $\eta < 1$, $\hat{\rho}$ is strictly increasing, so the set $\{T_1 : \rho > \hat{\rho}(T_1)\}$ is an initial segment of $[0, T_2]$. On it $\partial n/\partial T_1 > 0$, and after it $\partial n/\partial T_1 < 0$: n is single-peaked. The segment is all of $[0, T_2]$ if $\rho \geq \hat{\rho}(T_2)$, empty if $\rho \leq \rho_{\text{lo}}$, and otherwise ends at the unique T_1^* with $\hat{\rho}(T_1^*) = \rho$. Because $\hat{\rho}$ is strictly increasing, $T_1^* = \hat{\rho}^{-1}(\rho)$ rises with ρ.

(iii) If $\eta > 1$, $\hat{\rho}$ is strictly decreasing, so $\{T_1 : \rho > \hat{\rho}(T_1)\}$ is a terminal segment. $\partial n/\partial T_1 < 0$ before it and > 0 on it: n is single-troughed. The trough is interior when $\hat{\rho}(T_2) < \rho < \rho_{\text{lo}}$, at the unique $T_1^{\min}$ with $\hat{\rho}(T_1^{\min}) = \rho$, and $T_1^{\min} = \hat{\rho}^{-1}(\rho)$ falls with ρ.

Step 5: the case $\rho = 1$. Then $B = a > 0$ for every T_1 and every η, so n is strictly increasing. □

Proof of Proposition 2. Step 1: how h enters. By Lemma 1, π_S does not depend on h, while $\pi_X = mh\,e^{-1}$, so $\partial \ln \pi_X / \partial \ln h = 1$. From (7), $K = \pi_X A$ and $\Pi_{\text{fail}} = e^{-rT_1} K$ are proportional to π_X, so $\partial \ln K / \partial \ln h = \partial \ln \Pi_{\text{fail}} / \partial \ln h = 1$. From (6), only the second term of Π_{succ} involves π_X, so

$$\frac{\partial \ln \Pi_{\text{succ}}}{\partial \ln h} = \frac{\pi_X D(T_1, T_2)}{\Pi_{\text{succ}}(T_1)} = s(T_1).$$

Because $\Pi_{\text{succ}}(0) = \pi_X A$, $s(0) = 1$; because $D(T_2, T_2) = 0$, $s(T_2) = 0$; and s is strictly decreasing, since its numerator falls and its denominator rises with T_1.

Step 2: the two elasticities. For $c = rK\,\Pi_{\text{fail}}^{\eta-1}$, Step 1 gives $\partial \ln c / \partial \ln h = 1 + (\eta - 1) = \eta$. For $a = (\pi_S - \pi_X)\Pi_{\text{succ}}^{\eta-1}$, with π_S free of h,

$$\varepsilon_a = \frac{\partial \ln a}{\partial \ln h} = -\frac{\pi_X}{\pi_S - \pi_X} + (\eta - 1)\, s(T_1) = -\frac{\phi}{1-\phi} - (1-\eta)\, s(T_1),$$

using $\pi_X/(\pi_S - \pi_X) = \phi/(1-\phi)$.

Step 3: the cross-partial. In (8) the factor $\gamma_i \eta\, e^{-rT_1}$ does not depend on h, and $B = \rho\, a - (1-\rho)\, c$. Hence

$$\frac{\partial^2 n}{\partial T_1\, \partial h} = \gamma_i \eta\, e^{-rT_1} \Big[\rho \frac{\partial a}{\partial h} - (1-\rho) \frac{\partial c}{\partial h}\Big] = \frac{\gamma_i \eta\, e^{-rT_1}}{h} \Big[\rho\, a\, \varepsilon_a - (1-\rho)\, c\, \eta\Big],$$

since $\partial a/\partial h = a\,\varepsilon_a/h$ and $\partial c/\partial h = c\,\eta/h$ by Step 2.

Step 4: the signs. The gate term $-(1-\rho)\,c\,\eta$ is non-positive, and strictly negative for $\rho < 1$. If $\eta \leq 1$, both parts of ε_a are non-positive and the first is strictly negative, so $\rho\,a\,\varepsilon_a \leq 0$, strictly for $\rho > 0$. The bracket is therefore negative for every $\rho \in [0,1]$. If $\eta > 1$, then $\varepsilon_a < 0$ if and only if $(\eta-1)\,s(T_1) < \phi/(1-\phi)$. Because $s \leq 1$, this holds for every T_1 when $\eta - 1 < \phi/(1-\phi)$, that is, when $\eta < 1/(1-\phi)$, and the bracket is again negative.

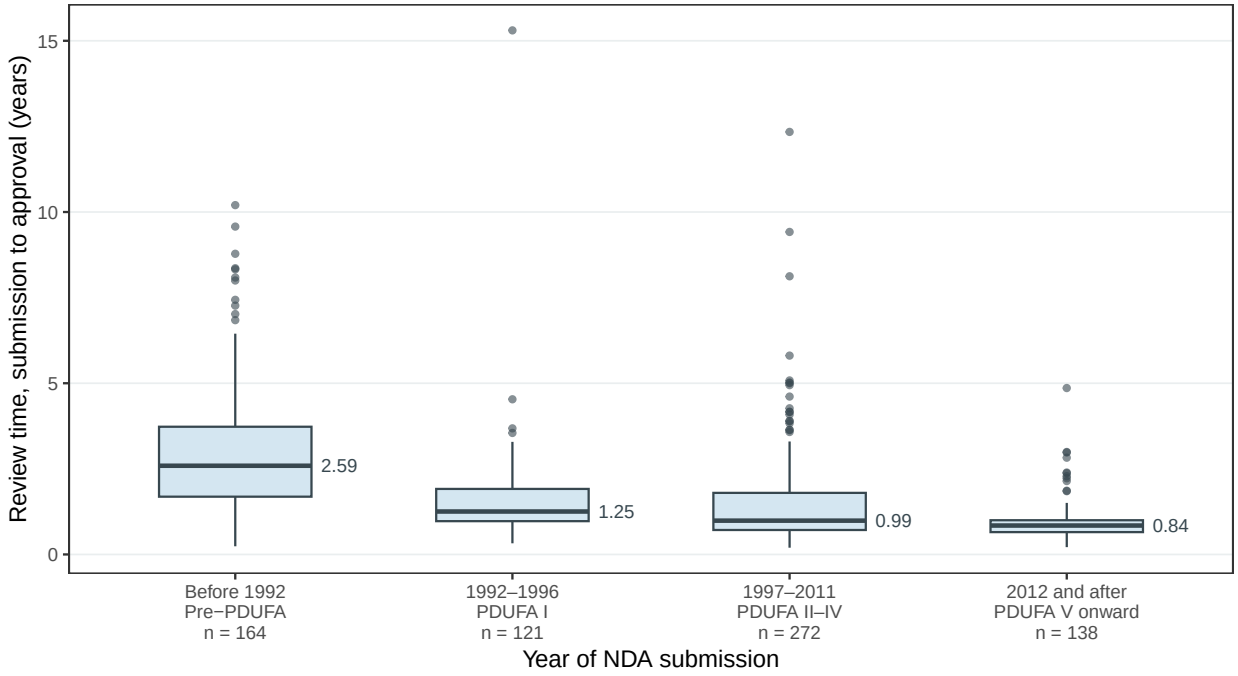
Step 5: the benchmark. Under $h > \Delta$ the developer is shielded throughout, so $\Pi_{\text{succ}} = \pi^* A$ and $\Pi_{\text{fail}} = e^{-rT_1}\pi^* A$ by Section 2.2. Neither involves π_X, so h does not enter (5), and the cross-partial is zero. □

Proof of Proposition 3. The originator needs no licence, so all of its uses lie in the licensed branch and $n_O = G_i(\Pi_{\text{succ}}) = \gamma_i\,\Pi_{\text{succ}}^{\eta}$. Differentiating, $\partial n_O/\partial T_1 = \gamma_i \eta\,\Pi_{\text{succ}}^{\eta-1}\,\partial\Pi_{\text{succ}}/\partial T_1 = \gamma_i \eta\, e^{-rT_1}(\pi_S - \pi_X)\Pi_{\text{succ}}^{\eta-1}$, which is positive because $\pi_S > \pi_X$ by Lemma 2 and $\eta > 0$. The entrant's count is (3), so $\partial n_E/\partial T_1$ is (8), whose sign is that of B; by Step 1 of the proof of Proposition 1, $B > 0$ if and only if $\rho > \hat{\rho}(T_1)$. In the benchmark the developer's own patent excludes all imitation, so a developer in the licensed branch earns π^* on $[0, T_2]$ whatever T_1: $\Pi_{\text{succ}} = \pi^* A$ is constant, hence $\partial n_O/\partial T_1 = 0$, and $\partial n_E/\partial T_1 \leq 0$ is Corollary 1. □

B Additional Figures and Tables

This appendix reports the distribution of FDA review time, the estimates behind Figure 6 and Figure 7, and the first stages of the instrumental-variable regressions.

Figure B.1: FDA review time by approval year



Notes: One box per group, for the drugs in the analysis sample. Review time runs from NDA submission to approval. The line in each box is the median, the box spans the interquartile range, and the whiskers extend to 1.5 times the interquartile range; points beyond them are individual drugs. Each group mark the enactment of PDUFA I (1992), PDUFA II (1997) and PDUFA V (2012).

Table B.1: First Stage: Market Exclusivity on FDA Review Time

	Market exclusivity (years)
FDA review time (years)	-0.543***
	(0.099)
Years on market	0.192***
	(0.020)
Patents at approval	0.064*
	(0.034)
Orphan drug	0.828*
	(0.441)
Priority review	0.078
	(0.346)
Instrument F	30.07
R-squared	0.148
Observations	695

Notes: Dependent variable: market exclusivity (years). This first stage is shared by columns 3 and 4 of Table 5. Instrument F is the robust partial F on review time. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.2: Functional Form of the Effect of Exclusivity

	Quadratic	Spline
Repurposing events (count)		
Exclusivity (years)	1.746***	
	(0.649)	
Exclusivity squared	-0.058**	
	(0.025)	
Slope, below 11 years		0.524***
		(0.129)
Slope, 11-16 years		0.233**
		(0.119)
Slope, above 16 years		-0.110
		(0.140)
First-stage F	25.6	25.6
Observations	695	695

Notes: Dependent variable: post-approval repurposing events per drug. Both columns are control-function Poisson regressions with exclusivity instrumented by FDA review time; the control function adds the first-stage residual, its square and its interaction with exclusivity. The spline has knots at 11 and 16 years and reports the slope of each segment. Controls as in Table 5, not reported. Bootstrap standard errors over both stages (999 replications). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.3: Exclusivity Response by Product Form

	Dependent variable: repurposing events (count)		
	Same product	New dosage form	Drug combination
Market exclusivity (years)	0.253***	0.339*	0.799
	(0.096)	(0.177)	(0.526)
Difference vs same product		0.085	0.545
		(0.153)	(0.513)
Years on market	-0.018	-0.009	-0.139
	(0.018)	(0.033)	(0.100)
Patents at approval	0.049***	0.019	0.011
	(0.015)	(0.027)	(0.069)
Orphan drug	-0.265	-0.306	-1.865
	(0.194)	(0.350)	(1.656)
Priority review	0.045	0.210	-0.199
	(0.163)	(0.310)	(0.826)
Drugs	695	695	695

Notes: Dependent variable: post-approval repurposing events per drug and product form. Same-product events are the benchmark: each other form is stacked against them in a control-function Poisson in which every coefficient is form-specific (Section 5.5), and the difference row is that form's response minus the same-product response. The two differences come from separate stacks that share the same-product arm, so they should not be pooled. Exclusivity is instrumented by FDA review time; controls as in Table 5. Standard errors from a drug-level bootstrap over both stages (999 replications). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.4: Exclusivity Response: Rare and Non-Rare Indications

	Dependent variable: repurposing events (count)	
	Rare indications	Non-rare indications
Market exclusivity (years)	0.399**	0.281**
	(0.188)	(0.118)
Difference (non-rare - rare)		-0.118
		(0.197)
Years on market	-0.057	-0.011
	(0.035)	(0.022)
Patents at approval	0.036	0.036*
	(0.028)	(0.018)
Orphan drug	0.958**	-1.204***
	(0.413)	(0.289)
Priority review	0.651*	-0.070
	(0.332)	(0.204)
Drugs	695	695

Notes: Dependent variable: post-approval repurposing events per drug and indication type (rare: orphan drug exclusivity; non-rare: all others). Both columns come from one stacked control-function Poisson in which every coefficient is type-specific (Section 5.5); the difference row is the non-rare minus the rare response. Exclusivity is instrumented by FDA review time; controls as in Table 5. Standard errors from a drug-level cluster bootstrap over both stages (999 replications). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.5: Exclusivity Response: Originators and Entrants

	Dependent variable: repurposing events (count)	
	Originator filings	Entrant filings
Market exclusivity (years)	0.541*	0.225**
	(0.320)	(0.101)
Difference (originator - entrant)	0.316	
	(0.327)	
Years on market	-0.107*	0.024
	(0.059)	(0.020)
Patents at approval	0.012	0.054**
	(0.036)	(0.023)
Orphan drug	-0.635	-0.236
	(0.473)	(0.253)
Priority review	-0.051	0.088
	(0.388)	(0.210)
One-sided p-value	0.045	0.013
Drugs	695	695

Notes: Dependent variable: post-approval repurposing events per drug, split by whether the filing firm owns the drug (originator) or not (entrant). Both columns come from one stacked control-function Poisson in which every coefficient is firm-type-specific (Section 5.5); the difference row is the originator minus the entrant response. The one-sided p-value tests a zero response (originators) or a non-positive response (entrants) against a positive one. Exclusivity is instrumented by FDA review time; controls as in Table 5. Standard errors from a drug-level cluster bootstrap over both stages (999 replications). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.6: First Stages: Review Time and the Priority-to-IND Gap

	All molecules	Molecules with an IND date	
	Review time	Priority to IND	Both
FDA review time (years)	-0.543***		-0.832***
	(0.099)		(0.192)
Patent priority to IND (years)		-0.300***	-0.305***
		(0.060)	(0.060)
Years on market	0.192***	0.103***	0.151***
	(0.020)	(0.028)	(0.027)
Patents at approval	0.064*	-0.173***	-0.166***
	(0.034)	(0.060)	(0.059)
Orphan drug	0.828*	0.283	0.344
	(0.441)	(0.695)	(0.684)
Priority review	0.078	0.536	-0.284
	(0.346)	(0.504)	(0.568)
Instrument F	30.07	24.82	20.26
R-squared	0.148	0.204	0.264
Observations	695	220	220

Notes: Dependent variable: market exclusivity (years). OLS first stages of Table 6; columns 2 and 3 are restricted to molecules with an IND date. Instrument F is the robust partial F on the excluded instrument, joint in column 3. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C The Observation Window

Table C.1 tests sensitivity to how I control for the observation window. The outcome counts repurposing events through 2024, giving earlier approvals more time to accumulate events. Review time correlates with this window (correlation 0.295): longer pre-PDUFA reviews tend to coincide with earlier approvals and longer windows. Inadequate window controls could therefore violate the exclusion restriction by linking review time to repurposing through observation length rather than exclusivity. I hold the instrument and other controls fixed while varying the window specification.

Columns 1–4 use different window controls. Column 1, the main specification, includes years on market linearly. Column 2 replaces this control with log years on market as an exposure term, fixing its coefficient at one. This assumes that expected events are proportional to observation length. Column 3 retains the exposure term and adds years on market linearly to relax proportionality. Column 4 includes years on market and its square to allow a nonlinear relationship. Column 5 retains the linear control but restricts the sample to drugs approved by 2014, ensuring at least ten years of observation.

Exclusivity has a positive effect in columns 1, 3, 4 and 5, significant at the 5 percent level or better. Coefficients range from 0.226 to 0.291, implying 25 to 34 percent more repurposing events per additional year. These results hold across window specifications. Columns 3 and 4 give slightly smaller coefficients than column 1, suggesting that the main estimate may be near the upper end. Column 5 restricts observation to at least ten years and gives a similar result, suggesting that recently approved drugs do not drive the effect. Column 2 gives a larger coefficient of 0.590, but its standard error of 1.042 makes it insignificant. Its proportionality assumption is inconsistent with observed event rates: annual events fall from about 0.12 for drugs observed ten years or less to about 0.05 for those observed more than thirty.

Table C.1: Treatment of the Observation Window

	Dependent variable: repurposing events (count)				
	Linear	Exposure	Exposure + linear	Quadratic	At least 10 years
Market exclusivity (years)	0.291***	0.590	0.247**	0.246**	0.226**
	(0.106)	(1.042)	(0.098)	(0.096)	(0.090)
Years on market	-0.018		-0.063***	0.053	-0.020
	(0.020)		(0.019)	(0.039)	(0.019)
Years on market, squared				-0.001*	
				(0.001)	
Patents at approval	0.040**	0.138	0.043***	0.044***	0.052***
	(0.017)	(0.104)	(0.015)	(0.016)	(0.016)
Orphan drug	-0.348	-0.002	-0.301	-0.299	-0.401*
	(0.215)	(0.507)	(0.198)	(0.194)	(0.226)
Priority review	0.071	-0.187	0.115	0.113	0.080
	(0.186)	(0.932)	(0.176)	(0.175)	(0.182)
First-stage F	30.07	10.23	30.07	29.47	29.81
Control-function t	-2.77	-0.58	-2.53	-2.57	-2.52
Observations	695	695	695	695	593

Notes: Dependent variable: post-approval repurposing events per drug through 2024. Control-function Poisson with exclusivity instrumented by FDA review time. Years on market enters linearly (column 1), only through a log exposure offset (2), through the offset and linearly (3), as a quadratic (4), and linearly on drugs approved by 2014 (5); the linear and quadratic terms enter both stages. Other controls as in Table 5. Bootstrap standard errors over both stages (999 replications). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.